Symmetry-Selected Nonlinear Thermal Pumping in Periodically Heated Mixed-Convection Channels

BENIAICHE Ahmed

Laboratory of Fluid Mechanics, Ecole Militaire Polytechnique, Algiers, Algeria

Email: beniaiche_ahmed25@yahoo.fr

Abstract

Nonlinear redistribution of wall heat flux in a two-dimensional laminar mixed-convection channel with spatially periodic wall-temperature modulation of unequal amplitude is governed by an exact symmetry selection rule, while the magnitude and the direction of the redistribution are set by the coupled thermal–hydrodynamic dynamics. Decomposing the first-order thermal response into symmetric and anti-symmetric Fourier components, we prove that their mixed quadratic interaction is odd across the channel, so that the second-order mean wall-flux corrections satisfy $Nu_{2,0}(+1) = -Nu_{2,0}(-1)$ identically, whereas symmetry-preserving self-interactions produce no directional redistribution at the same order. The parity law defines a thermal-pumping coefficient Π_T that obeys the exact Green-function identity $\Pi_T = -(Pe/2)\int_{-1}^{1} Y\mathcal{E}_{as}\,dY$, from which the linearity $\Pi_T = GrA(k, Re, Pr)$, the long-wavelength scaling $A = O(k^2)$, and the corollary that the zero of Π_T is independent of Gr all follow. An independent fully nonlinear DNS converges to $\Pi_T^{\mathrm{DNS}} = -4.6763 \times 10^{-2}$ for $(k, Pe, Re, Pr, Gr) = (0.8, 7, 10, 0.7, 100)$, within 1.52% of the perturbative prediction, reproduces the symmetry-selected reversal upon interchange of the two wall amplitudes, and confirms that the reversal point in Re does not move when Gr is varied over 25–200. At fixed Pe the reversal location $Re_c(k)$ is mapped perturbatively over seven wavenumbers, from $Re_c = 79.08$ at $k = 0.4$ to $Re_c = 8.54$ at $k = 2.0$. A thirteen-case DNS campaign then confirms the reversal nonlinearly on both sides of the crossing at four wavenumbers, all eight bracketed cases returning the predicted sign, reproduces the parity cancellation to 3.1×10^{-18} in the wall-flux difference, measures the quadrature coefficient to 1.38%, and extends the Grashof-independence of the reversal to the nonlinear regime: the measured Grashof ratios of $q_+ - q_-$ lie within 3.7×10^{-4} of the exact factor of four and the DNS-minus-perturbative discrepancy is itself proportional to Gr, so the discrepancy rescales the coefficient without moving its zero. The same campaign falsifies part of

its own pre-registered error model—five of the twelve non-degenerate cases exceed the bound, with a deviation independent of the forcing amplitude and growing with the wavenumber—and the residual is analysed as a remainder whose origin, truncation or spatial discretisation, is not separated. The parity law is shown to extend to phase-shifted forcing through a second, independent pumping coefficient $\Pi_T^{\perp}$ that enters with the quadrature product $\mathrm{Im}(\varepsilon_a \varepsilon_s^*)$ and reverses the pumping direction on its own. Symmetry therefore selects the *structure* of the nonlinear redistribution, whereas buoyancy-mediated dynamics select its amplitude, its sign, and the control parameter that governs it.

Keywords: mixed convection; streamwise-periodic wall heating; thermal pumping; symmetry selection rule; weakly nonlinear expansion; Reynolds-number reversal; direct numerical simulation; numerical verification

Nomenclature

Roman Symbols	
$A(k, Re, Pr)$	Normalized pumping response, $A = \Pi_T / Gr$ [-]
H	Channel half-width [m]
k	Dimensionless modulation wavenumber [-]
k^*	Zero of $\Pi_T(k, Pe)$ when an interior crossing exists [-]
Nu	Nusselt number [-]
p	Pressure [Pa]
Pe	Péclet number, $Pe = RePr$ [-]
Pr	Prandtl number, $Pr = \nu/\alpha$ [-]
Gr	Grashof number, $Gr = g\beta\Delta T H^3/\nu^2$ [-]
Re	Reynolds number, $Re = U_0 H/\nu$ [-]
U_0	Reference mean velocity used for nondimensionalization [m/s]
X, Y	Dimensionless streamwise and transverse coordinates [-]
Greek Symbols	
α	Thermal diffusivity [m^2/s]
β	Thermal expansion coefficient [K^{-1}]
$\varepsilon_1, \varepsilon_2$	Dimensionless wall-modulation amplitudes [-]
$\varepsilon_s, \varepsilon_a$	Symmetric and antisymmetric amplitudes [-]
θ	Dimensionless temperature perturbation [-]
ν	Kinematic viscosity [m^2/s]
Π_T	Normalized nonlinear thermal pumping coefficient [-]
ϕ	Phase shift between wall modulations [rad]
Abbreviations	
BC	Boundary condition
BVP	Boundary-value problem
IVP	Initial-value problem
RK	Runge–Kutta

1 Introduction

Two mechanisms are deliberately separated throughout this paper. Reflection *symmetry* selects the *structure* of the nonlinear redistribution of the wall heat flux: the second-order mean corrections at the two walls are exactly equal in magnitude and opposite in sign. The thermal–hydrodynamic *dynamics* select the *amplitude and the sign* of that redistribution, and at fixed Péclet number the Reynolds number controls the direction of the pumping. Every result below is assigned to one of these two categories, and the central claim of the paper is that the division is exact rather than rhetorical: the first statement holds as a theorem under stated assumptions, whereas the second requires quantitative calculation and can even reverse.

Mixed convection in vertical channels is a classical problem in which imposed shear and buoyancy interact to modify the velocity, temperature, and heat-transfer fields. Early analyses established the fully developed theory and the possibility of buoyancy-induced flow reversal in vertical parallel- plate channels Aung and Worku (1986b,a); Hamadah and Wirtz (1991), while subsequent work considered asymmetric wall heating and its effect on the hydrodynamic and thermal response Aung and Worku (1987); Barletta (1998, 1999). These studies provide the mixed-convection foundation for the present analysis.

Spatially periodic thermal forcing introduces a separate mechanism: the wall temperature varies along the streamwise direction and generates thermal modes that interact with the base flow and, through buoyancy, with perturbation velocities. Periodically heated channels have been investigated both for natural convection and mixed convection, including vertical-channel configurations with spatially periodic wall temperatures Dritselis et al. (2013); Hossain and Floryan (2015). The present problem is distinct in that the central observable is not an instability threshold or a global heat-transfer enhancement, but a second-order redistribution of the mean wall heat flux caused by deliberate breaking of the reflection symmetry between the two walls.

The word *pumping* requires a qualification, because it is used in the literature for a different device. Thermal-wave pumping produces a net mean flow by means of a temperature wave travelling along the boundary Hossain and Floryan (2023); the present rectified quantity is not a propulsive flux but the difference between the two wall heat fluxes, i.e. a redistribution of heat across the channel at strictly fixed bulk flow rate. The mechanisms are also unrelated: there the asymmetry is imposed by a propagating phase, here it is imposed by the parity of a standing quadratic interaction between a symmetric and an antisymmetric thermal mode.

Weakly nonlinear expansions are a natural tool for identifying such rectified responses. In hydrodynamic stability theory, nonlinear interactions between Fourier modes and the associated mean-flow corrections form a standard framework Stuart (1960); symmetry and mode selection are likewise central to the broader theory of pattern-forming systems Cross

and Hohenberg (1993). Here, however, the reference state is not studied as an instability threshold. The weakly nonlinear expansion is used to expose an exact symmetry selection rule embedded in the steady forced transport problem.

For identical upper and lower wall modulations, $\varepsilon_1 = \varepsilon_2$, the leading second-order wall-flux redistribution is symmetry-forbidden. The corresponding nonlinear forcing is even in the transverse coordinate and cannot generate a persistent difference between the two inward wall heat fluxes. The symmetric configuration therefore supplies a precise baseline against which the effect of structural asymmetry can be isolated.

The present study asks what happens when $\varepsilon_1 \neq \varepsilon_2$. We answer this question at two levels. First, reflection symmetry and uniqueness produce an exact selection rule: the mixed symmetric–antisymmetric interaction is odd in the transverse coordinate, and therefore its second-order wall-flux corrections are equal in magnitude and opposite in sign. Second, the amplitude and sign of this redistribution are determined by the coupled thermal and hydrodynamic response and hence remain parameter dependent. This distinction is central: wall-flux antisymmetry is exact under the stated assumptions, whereas nonzero pumping, its $k \to 0$ scaling, and the location of its zero crossing are dynamical results requiring quantitative calculation.

The contributions of the paper are therefore threefold.

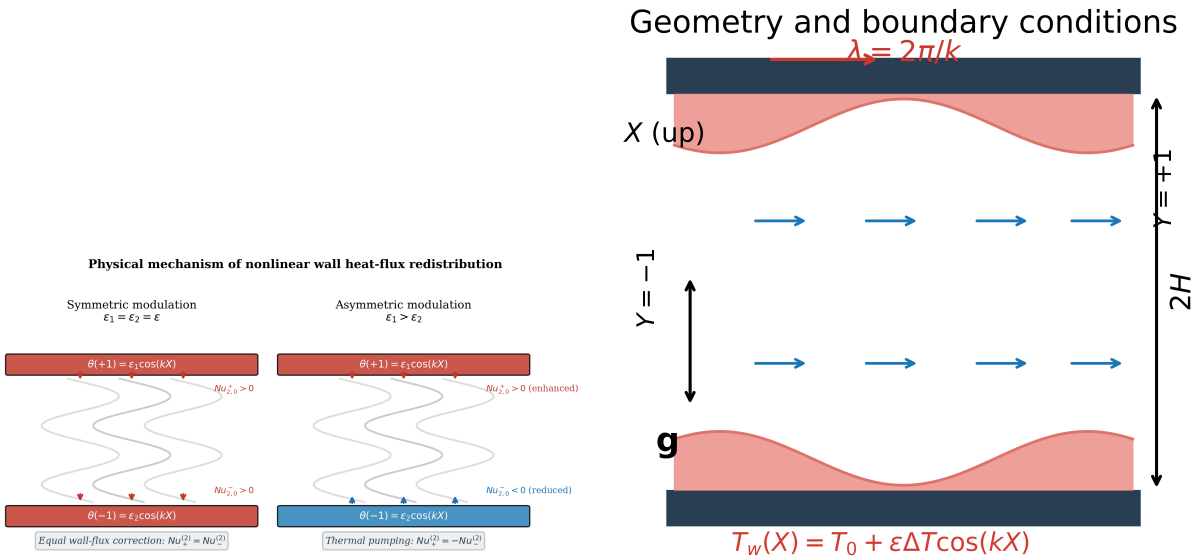
1. An **exact symmetry-selection theorem** (Theorem 1): the mixed symmetric–antisymmetric second-order forcing is odd, hence $Nu_{2,0}(+1) = -Nu_{2,0}(-1)$ for every admissible (Re, Pr, Gr, k), and pure symmetric or pure antisymmetric heating produce no directional redistribution at this order.

2. An **exact two-coefficient phase law** (Proposition 2): for phase-shifted wall modulation the rectified wall-flux response is governed by the real and imaginary parts of the product $\varepsilon_a \varepsilon_s^*$, each multiplying an independent pumping coefficient. For equal wall amplitudes the dependence on the phase is exactly $\sin\phi$, with no higher harmonic, and the quadrature coefficient $\Pi_T^{\perp}$ has the *opposite sign* to the published in-phase coefficient Π_T.

3. A **dynamical mapping of the reversal**, validated by fully nonlinear DNS: the parity cancellation itself is reproduced in DNS to 3.1×10^{-18} in the wall-flux difference, which is 2.7×10^{-11} of the flux measured; the reversal point Re_c at fixed Pe is independent of Gr to 1.2×10^{-10} perturbatively and keeps its sign across a Grashof pair in DNS; the locus is mapped as a function of k and confirmed on both sides of four of its crossings; and a term-wise phase analysis identifies which nonlinear transport contribution loses the balance as Re increases.

What is deliberately *not* claimed is universality: the results are two-dimensional, laminar, and restricted to the parameter domain where the second-order expansion is quantitatively controlled.

Relation to the companion studies. This paper is the third and symmetry-selecting member of a series on the same canonical configuration. The first study treats the *symmetric* wall modulation and develops the exact integral framework for the first-order complex thermal amplitude, including the non-monotonic k-dependence of the mode-coupling coefficient. The second study treats the same symmetric problem with a corrected modal decomposition and a fourth-order hydrodynamic boundary-value problem, and establishes that $Nu_{2,0} = 0$ in that configuration, together with the three-regime map of the first-order response. The present study changes the object of interest: it is the asymmetric configuration, the exact parity selection rule that follows from it, and the dynamical consequences of that rule. Only the shared canonical geometry and the standard nondimensional Navier–Stokes–Boussinesq statement appear again; no derivation, table or figure of the companion studies is reproduced here. Neither companion manuscript has yet appeared in print or on a preprint server, so neither carries a citable identifier; they are therefore described in the text rather than cited in the reference list, and are supplied to the editor on request. Fig. 1 shows the configuration and the two parity classes of wall modulation it makes accessible, and Fig. 2 traces the chain from wall symmetry to the admissible second-order response.



(a) Mechanism: symmetric versus unequal wall amplitudes.

(b) Geometry, base flow and nondimensionalization.

Figure 1: Physical schematic of the wall heat-flux redistribution mechanism and the configuration on which it acts. The symmetric limit suppresses the antisymmetric wall-flux correction, whereas unequal wall-modulation amplitudes activate the mixed symmetric–antisymmetric interaction responsible for the second-order redistribution. The vertical channel has walls at $Y = \pm 1$, half-width H, base Poiseuille flow $U_0(Y) = \frac{3}{2}(1 - Y^2)$, gravity in the vertical direction, and periodically modulated wall temperatures with amplitudes ε_1 and ε_2.

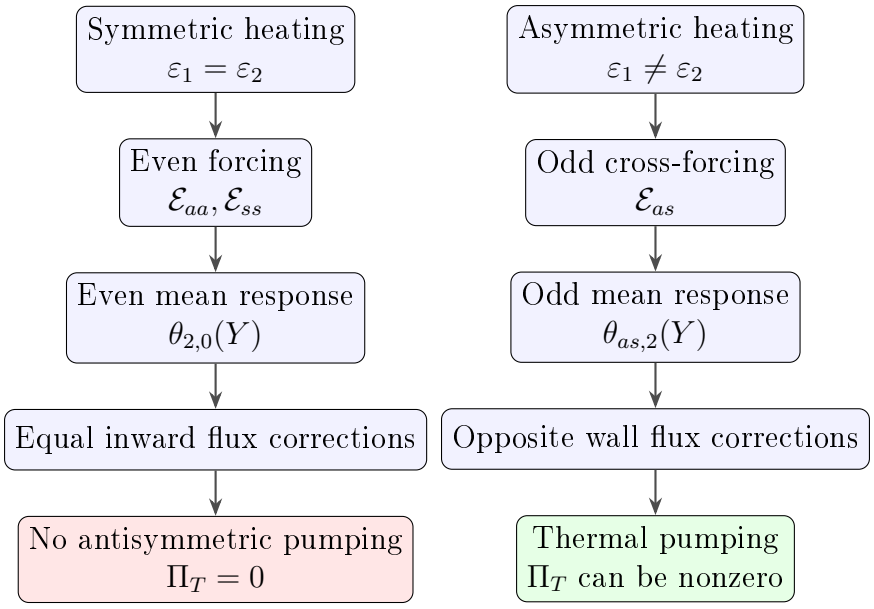


Figure 2: Logical chain connecting wall-temperature symmetry to the nonlinear wall-flux response.

2 Formulation and Base State

A laminar, incompressible Newtonian fluid flows steadily through a vertical parallel-plate channel with walls at $Y = \pm 1$.[1] The dimensional half-width is H, and the reference velocity U_0 is the bulk velocity used for nondimensionalization. The X axis is vertical and gravity acts in the negative X direction. After removal of the hydrostatic pressure contribution, the dimensionless Boussinesq system is

$$\partial_X U + \partial_Y V = 0, \tag{1}$$

$$U\partial_X U + V\partial_Y U = -\partial_X P + \frac{1}{Re}\nabla^2 U + \frac{Gr}{Re^2}\theta, \tag{2}$$

$$U\partial_X V + V\partial_Y V = -\partial_Y P + \frac{1}{Re}\nabla^2 V, \tag{3}$$

$$U\partial_X \theta + V\partial_Y \theta = \frac{1}{RePr}\nabla^2 \theta, \tag{4}$$

where $Re = U_0 H/\nu$, $Pr = \nu/\alpha$, $Pe = RePr$, and $Gr = g\beta\Delta T H^3/\nu^2$. The applicability of the Boussinesq approximation is assumed in the usual small-density-variation regime Gray and Giorgini (1976).

[1]The Boussinesq formulation, the plane-Poiseuille base state, the nondimensionalization and the first-order thermal problem are reproduced here for completeness. A word-level duplication scan of this manuscript against the two companion studies described in the Introduction finds three shared twelve-word shingles with the first and none with the second, and all three lie in the author-and-affiliation block; no scientific sentence is carried over from either companion. The larger overlaps in the working archive—1.8% and 0.6% shared shingles—are with earlier drafts of this same manuscript, from which it differs by the second-order parity argument of Section 3 and everything built on it; the drafts share with it only the drawing source of the parity chain diagram, the nomenclature table and the formulation block above.

The unperturbed hydrodynamic state is the fully developed Poiseuille flow

$$U_0(Y) = \frac{3}{2}(1 - Y^2), \qquad V_0 = 0, \tag{5}$$

whose cross-sectional mean is unity. The zero-order thermal state is taken as the reference state, so that the imposed wall modulation is

$$\theta(X, +1) = \varepsilon_1 \cos(kX), \tag{6}$$

$$\theta(X, -1) = \varepsilon_2 \cos(kX). \tag{7}$$

No-slip conditions apply at both walls and the solution is periodic in X with period $2\pi/k$.

The wall amplitudes are decomposed as

$$\varepsilon_s = \frac{\varepsilon_1 + \varepsilon_2}{2}, \qquad \varepsilon_a = \frac{\varepsilon_1 - \varepsilon_2}{2}, \tag{8}$$

so that $\varepsilon_1 = \varepsilon_s + \varepsilon_a$ and $\varepsilon_2 = \varepsilon_s - \varepsilon_a$.

For the first-order fields we use

$$\theta_1^{(j)}(X, Y) = \Re\{\Theta^{(j)}(Y) e^{-ikX}\}, \tag{9}$$

$$V_1^{(j)}(X, Y) = \Re\{\widehat{V}^{(j)}(Y) e^{-ikX}\}, \tag{10}$$

with $j \in \{s, a\}$ and $\Theta^{(j)} = \theta_c^{(j)} + i\theta_s^{(j)}$. A streamfunction may be introduced according to

$$U_1^{(j)} = \partial_Y \psi_1^{(j)}, \qquad V_1^{(j)} = -\partial_X \psi_1^{(j)}, \tag{11}$$

so incompressibility is identically satisfied. With the e^{-ikX} convention, continuity gives

$$\boxed{\widehat{U}^{(j)} = -\frac{i}{k}(\widehat{V}^{(j)})'.} \tag{12}$$

The weakly nonlinear expansion is organized as

$$\theta = \varepsilon_s \theta_1^{(s)} + \varepsilon_a \theta_1^{(a)} + \varepsilon_s^2 \theta_{ss,2} + \varepsilon_a^2 \theta_{aa,2} + \varepsilon_a \varepsilon_s \theta_{as,2} + O(\varepsilon^3), \tag{13}$$

$$(U, V) = (U_0, 0) + \varepsilon_s (U_1^{(s)}, V_1^{(s)}) + \varepsilon_a (U_1^{(a)}, V_1^{(a)}) + O(\varepsilon^2). \tag{14}$$

2.1 First-order thermal problems

The first-order thermal amplitudes satisfy

$$\mathcal{L}_T \Theta^{(j)} \equiv (D^2 - k^2)\Theta^{(j)} + ikPeU_0(Y)\Theta^{(j)} = 0, \qquad D \equiv \frac{d}{dY}, \tag{15}$$

with

$$\Theta^{(s)}(-1) = \Theta^{(s)}(+1) = 1, \tag{16}$$

$$\Theta^{(a)}(-1) = -1, \qquad \Theta^{(a)}(+1) = 1. \tag{17}$$

Thus the thermal modes depend on the thermal control combination Pe together with k. For completeness we note that problem (15) with symmetric wall data is the first-order thermal problem analysed in the companion integral-identity study of this series; it is reproduced here because the present argument needs both parities simultaneously, and none of the results quoted below is taken from that study.

2.2 First-order hydrodynamic response

Let $\mathcal{L} = D^2 - k^2$. Taking the curl of the linearized momentum equations and eliminating pressure gives

$$\mathcal{L}^2\widehat{V}^{(j)} + ikRe\left[U_0\mathcal{L}\widehat{V}^{(j)} - U_0''\widehat{V}^{(j)}\right] + i\frac{kGr}{Re}\big(\Theta^{(j)}\big)' = 0. \tag{18}$$

with

$$\widehat{V}^{(j)}(\pm 1) = 0, \qquad (\widehat{V}^{(j)})'(\pm 1) = 0. \tag{19}$$

The corresponding linear vorticity relation is

$$U_0\partial_X\omega_1 - V_1U_0'' = \frac{1}{Re}\nabla^2\omega_1 - \frac{Gr}{Re^2}\partial_Y\theta_1. \tag{20}$$

Because the thermal fields are independent of Gr and the hydrodynamic operator is linear, at fixed (k, Re, Pr),

$$\widehat{V}^{(j)} = Gr\,\widetilde{V}^{(j)}, \qquad \widehat{U}^{(j)} = Gr\,\widetilde{U}^{(j)}. \tag{21}$$

3 The Symmetry Law: Odd Parity of the Cross-Interaction

3.1 Reflection parity of the first-order thermal modes

Let $\mathcal{R}[f](Y) = f(-Y)$. Since U_0 is even, $\mathcal{R}$ commutes with the thermal operator. Uniqueness therefore gives

$$\Theta^{(s)}(-Y) = \Theta^{(s)}(Y), \qquad \Theta^{(a)}(-Y) = -\Theta^{(a)}(Y). \tag{22}$$

Hence $\Theta^{(s)}$ is even and $(\Theta^{(s)})'$ odd, while $\Theta^{(a)}$ is odd and $(\Theta^{(a)})'$ even.

3.2 Reflection parity of the first-order hydrodynamic modes

Since U_0 and U_0'' are even, the hydrodynamic operator is also reflection invariant. The forcing in Eq. (18) is odd for the symmetric thermal mode and even for the antisymmetric one. Uniqueness of the homogeneous-boundary hydrodynamic BVP gives

$$\widehat{V}^{(s)}(-Y) = -\widehat{V}^{(s)}(Y), \qquad \widehat{V}^{(a)}(-Y) = \widehat{V}^{(a)}(Y), \tag{23}$$

and Eq. (12) gives

$$\widehat{U}^{(s)}(-Y) = \widehat{U}^{(s)}(Y), \qquad \widehat{U}^{(a)}(-Y) = -\widehat{U}^{(a)}(Y). \tag{24}$$

3.3 Explicit quadratic cross-interaction

The streamwise-averaged second-order energy equation is

$$\theta_{2,0}'' = Pe\left[\varepsilon_a^2\mathcal{E}_{aa} + \varepsilon_s^2\mathcal{E}_{ss} + \varepsilon_a\varepsilon_s\mathcal{E}_{as}\right], \qquad \theta_{2,0}(\pm 1) = 0, \tag{25}$$

where

$$\mathcal{E}_{jj}(Y) = \left\langle U_1^{(j)}\partial_X\theta_1^{(j)} + V_1^{(j)}\partial_Y\theta_1^{(j)}\right\rangle_X, \qquad j \in \{s,a\}, \tag{26}$$

$$\mathcal{E}_{as}(Y) = \left\langle U_1^{(a)}\partial_X\theta_1^{(s)} + U_1^{(s)}\partial_X\theta_1^{(a)} + V_1^{(a)}\partial_Y\theta_1^{(s)} + V_1^{(s)}\partial_Y\theta_1^{(a)}\right\rangle_X. \tag{27}$$

The complex-amplitude form is

$$\begin{aligned}\mathcal{E}_{as}(Y) = \frac{1}{2}\Re\Big[&\widehat{U}^{(a)}\overline{-ik\Theta^{(s)}} + \widehat{U}^{(s)}\overline{-ik\Theta^{(a)}} \\ &+ \widehat{V}^{(a)}\overline{(\Theta^{(s)})'} + \widehat{V}^{(s)}\overline{(\Theta^{(a)})'}\Big].\end{aligned} \tag{28}$$

3.4 Second-order mean momentum and streaming current

The streamwise-averaged second-order momentum equation may be written as

$$\frac{1}{Re}U_{2,0}'' + G_2 = \mathcal{N}(Y) - \frac{Gr}{Re^2}\theta_{2,0}(Y), \tag{29}$$

where

$$\mathcal{N}(Y) = \langle U_1\partial_X U_1 + V_1\partial_Y U_1\rangle_X. \tag{30}$$

For fixed bulk flow,

$$U_{2,0}(\pm 1) = 0, \qquad \int_{-1}^{1} U_{2,0}(Y)\,dY = 0, \tag{31}$$

which determines the pressure-gradient correction G_2. The streaming calculation is therefore a separate second-order problem; the wall-flux symmetry theorem itself follows al-

ready from the first-order thermal and transverse-velocity parities.

The parity table is

	symmetric (s)	antisymmetric (a)
Θ	even	odd
Θ'	odd	even
$\widehat{V}$	odd	even
$\widehat{U}$	even	odd

Every term in Eq. (28) is therefore odd, so

$$\boxed{\mathcal{E}_{as}(-Y) = -\mathcal{E}_{as}(Y)}. \tag{32}$$

Also, $U_1\partial_X\theta_1 + V_1\partial_Y\theta_1 = \partial_X(U_1\theta_1) + \partial_Y(V_1\theta_1)$. Periodic averaging and $V_1(\pm 1) = 0$ yield

$$\int_{-1}^{1} \mathcal{E}_{as}(Y)\, dY = 0. \tag{33}$$

3.5 Solvability and wall-flux identities

For the mixed contribution,

$$\theta''_{as,2} = Pe\, \mathcal{E}_{as}, \qquad \theta_{as,2}(\pm 1) = 0. \tag{34}$$

Because the forcing is odd, uniqueness gives

$$\theta_{as,2}(-Y) = -\theta_{as,2}(Y), \qquad \theta'_{as,2}(-Y) = \theta'_{as,2}(Y). \tag{35}$$

With inward-normal wall fluxes

$$Nu_{2,0}(+1) = -\theta'_{as,2}(+1), \qquad Nu_{2,0}(-1) = +\theta'_{as,2}(-1), \tag{36}$$

we obtain

$$\boxed{Nu_{2,0}(+1) = -Nu_{2,0}(-1)}. \tag{37}$$

This is the exact symmetry theorem. It constrains the two-wall relation but does not imply a nonzero coefficient at every parameter value.

For the symmetry-preserving pure symmetric and pure antisymmetric cases, the second-order forcing is even, the mean temperature correction is even, and the two inward flux corrections are equal. The periodic energy balance gives zero net second-order heat input, so the equal corrections must vanish individually.

Theorem 1 (Nonlinear thermal pumping symmetry law)**.** *Assume a reflection-symmetric laminar base state, unique first-order thermal and hydrodynamic BVP solutions, spatially*

periodic harmonic wall forcing, and validity of the second-order perturbation expansion. Then the mixed symmetric–antisymmetric second-order contribution satisfies

$$Nu_{2,0}(+1) = -Nu_{2,0}(-1) \tag{38}$$

for every admissible (Re, Pr, Gr, k) *for which the stated perturbative expansion and the associated unique boundary-value problems are valid.*

Proof. The reflection parities of the first-order thermal and velocity modes are given by Eqs. (22)–(24). They make every term of the mixed quadratic forcing odd, hence $\mathcal{E}_{as}$ is odd. The unique Dirichlet solution of Eq. (34) is therefore odd and its derivative even. Applying the opposite inward normals at $Y = \pm 1$ gives Eq. (37). □

4 Exact Identities and Thermal Pumping Coefficient

4.1 Definition

The normalized transverse wall-flux redistribution is defined by

$$\Pi_T(k, Pe, Gr; Re, Pr) = \frac{Nu_{2,0}(+1) - Nu_{2,0}(-1)}{2\varepsilon_a\varepsilon_s} = \frac{Nu_{2,0}(+1)}{\varepsilon_a\varepsilon_s}. \tag{39}$$

With the present sign convention, $\Pi_T > 0$ corresponds to enhanced inward flux at the upper wall relative to the lower wall, while $\Pi_T < 0$ corresponds to the reverse redistribution.

4.2 Exact Green-function identity

For $\theta'' = f(Y)$ with homogeneous Dirichlet conditions,

$$\theta'(1) = \frac{1}{2}\int_{-1}^{1}(1+Y)f(Y)\,dY, \qquad \theta'(-1) = -\frac{1}{2}\int_{-1}^{1}(1-Y)f(Y)\,dY. \tag{40}$$

For the odd forcing $f = Pe\mathcal{E}_{as}$,

$$\boxed{\Pi_T = -\frac{Pe}{2}\int_{-1}^{1} Y\,\mathcal{E}_{as}(Y)\,dY.} \tag{41}$$

This identity is independent of the numerical method used for the mean second-order problem and is therefore an especially useful verification target.

Proposition 1 (Linearity in Grashof number)**.** *At fixed* (k, Re, Pr)*,*

$$\boxed{\Pi_T(k, Pe, Gr; Re, Pr) = GrA(k, Re, Pr).} \tag{42}$$

Proof. The thermal amplitudes are independent of Gr, whereas the hydrodynamic BVP is linear and its forcing is proportional to $Gr\Theta'$. Hence both $\widehat{V}$ and $\widehat{U}$ are proportional to Gr, so Eq. (28) gives $\mathcal{E}_{as} = Gr\widetilde{\mathcal{E}}_{as}$. Equation (41) then gives the asserted linearity. □

Corollary 1 (Grashof-independence of the reversal point)**.** *Because* $\Pi_T(k, Pe, Gr; Re, Pr) = GrA(k, Re, Pr)$ *with* $Gr > 0$*, the set of* (k, Pe, Re, Pr) *at which* Π_T *vanishes coincides exactly with the set at which* A *vanishes. The location of a pumping reversal is therefore independent of* Gr*: varying the buoyancy amplitude rescales the response without moving its zero or changing its sign. The same argument applies verbatim to the quadrature coefficient* $\Pi_T^{\perp}$ *of Proposition 2, since it is also proportional to* Gr *at fixed* (k, Re, Pr)*.*

Proof. Linearity of Proposition 1 gives $\Pi_T/Gr = A$ with A independent of Gr; since $Gr > 0$, $\Pi_T = 0 \iff A = 0$, and $\operatorname{sgn}\Pi_T = \operatorname{sgn} A$ for every Gr. The quadrature coefficient obeys the identical scaling because the hydrodynamic modes that enter its kernel are the same Gr-linear solutions. □

This corollary is a structural prediction rather than an asymptotic one, and it is directly testable. At $k = 0.8$ and $Pe = 7$ the reversal Reynolds number was located by Brent iteration on the authoritative perturbative solver for four Grashof numbers:

Gr	25	50	100	200
Re_c	30.838496520	30.838496520	30.838496520	30.838496520

(43)

the four values agreeing to the nine displayed digits, with an absolute spread of 1.2×10^{-10}. The corresponding normalized responses $A = \Pi_T/Gr$ are invariant across the same sweep to a worst-case relative spread of 7.3×10^{-11} at $Re = 25, 30, 35$ and to 1.0×10^{-13} at the reference $Re = 10$ (Table 7), i.e. to the convergence level of the boundary-value solver rather than to any modelling approximation.

4.3 Parameter dependence and Reynolds-number dependence

The normalized response $A = \Pi_T/Gr$ is, in general, a function of (k, Re, Pr). Although the thermal operator contains Re and Pr through the product $Pe = RePr$, the hydrodynamic operator retains explicit dependence on Re. Consequently there is no exact reduction to $A(k, Pe)$ in the general problem.

At fixed $Re = 10$, however, one may use Pe as the thermal control parameter by setting $Pr = Pe/Re$. A direct fixed-Pe test at $k = 0.8$ and $Gr = 100$ confirms that the resulting map is not universal with respect to Re. For $Pe = 7$, for example,

Re	5	10	20	35
$A = \Pi_T/Gr$	-1.5951×10^{-3}	-4.7485×10^{-4}	-6.4279×10^{-5}	6.7452×10^{-6}

(44)

so the Reynolds number remains an active hydrodynamic control parameter, and Table 1 gives the corresponding fixed-Pe map.

Table 1: Fixed-Pe dependence of $A = \Pi_T/Gr$ at $k = 0.8$, $Gr = 100$.

Pe	$Re = 5$	$Re = 10$	$Re = 20$	$Re = 35$
2	-4.5382×10^{-4}	-3.9078×10^{-5}	6.1165×10^{-5}	4.5851×10^{-5}
4	-1.2126×10^{-3}	-3.1884×10^{-4}	-1.3565×10^{-5}	2.4576×10^{-5}
7	-1.5951×10^{-3}	-4.7485×10^{-4}	-6.4279×10^{-5}	6.7452×10^{-6}
12	-1.6138×10^{-3}	-5.0178×10^{-4}	-8.5084×10^{-5}	-4.0374×10^{-6}
20	-1.5130×10^{-3}	-5.0141×10^{-4}	-1.0914×10^{-4}	-1.9812×10^{-5}

4.4 Long-wavelength limit

For $k \to 0$,

$$\Theta^{(s)}(Y) = 1 + ika_s(Y) + O(k^2), \qquad a_s'' = -PeU_0, \qquad a_s(\pm 1) = 0, \tag{45}$$

$$\Theta^{(a)}(Y) = Y + ika_a(Y) + O(k^2), \qquad a_a'' = -PeU_0 Y, \qquad a_a(\pm 1) = 0. \tag{46}$$

Thus $(\Theta^{(s)})' = O(k)$ whereas $(\Theta^{(a)})' = 1 + O(k)$. Equation (18) then gives

$$\widehat{V}^{(a)} = O(k), \qquad \widehat{U}^{(a)} = O(1), \qquad \widehat{V}^{(s)} = O(k^2), \qquad \widehat{U}^{(s)} = O(k), \tag{47}$$

so every cross-forcing contribution is $O(k^2)$ and

$$\boxed{A(k, Re, Pr) = A_2(Re, Pr)k^2 + o(k^2), \qquad k \to 0.} \tag{48}$$

The numerical sequence at $(Re, Pr, Gr) = (10, 0.7, 100)$ gives a log–log exponent $p = 1.9955$ with $R^2 = 1.0000$ over $0.002 \le k \le 0.03$, while Π_T/k^2 remains close to -2.2. The observed exponent is therefore consistent with the analytic k^2 law; Table 2 gives the fit over the tested interval.

Table 2: Long-wavelength fit at $(Re, Pr, Gr) = (10, 0.7, 100)$.

exponent p	prefactor	R^2	standard error	number of points
1.9955	2.1665	1.0000	0.00133	7

For large k, the temperature perturbations become increasingly localized near the two walls, with a generic geometric penetration scale proportional to k^{-1}. We use this only as a qualitative explanation for the decay of cross-wall coupling and do not infer a universal large-k exponent. The indicator $\Lambda = (kPe)^{-1/2}$ is introduced only as a scaling measure for the competition between streamwise advection and transverse diffusion.

5 Exact Phase Selection and the Quadrature Pumping Coefficient

5.1 Complex forcing amplitudes

Let the two wall modulations differ both in amplitude and in streamwise phase,

$$\theta(X,+1) = \varepsilon_1 \cos(kX), \qquad \theta(X,-1) = \varepsilon_2 \cos(kX+\phi), \tag{49}$$

with $\varepsilon_1, \varepsilon_2 \geq 0$ and $\phi \in [0, 2\pi)$. Writing every first-order field as $\Re\big[\widehat{(\cdot)}\, e^{-ikX}\big]$, the wall data are $\widehat{\theta}_1(+1) = \varepsilon_1$ and $\widehat{\theta}_1(-1) = \varepsilon_2 e^{-i\phi}$, and the symmetric/antisymmetric split becomes

$$A_s = \tfrac{1}{2}\big(\varepsilon_1 + \varepsilon_2 e^{-i\phi}\big), \qquad A_a = \tfrac{1}{2}\big(\varepsilon_1 - \varepsilon_2 e^{-i\phi}\big). \tag{50}$$

Because the first-order thermal and hydrodynamic problems are linear and invariant under streamwise translation, the response to a complex amplitude is the amplitude times the unit-amplitude response. Only the two real scalars

$$c = \Re\big(A_a A_s^*\big), \qquad d = \mathrm{Im}\big(A_a A_s^*\big) \tag{51}$$

survive the streamwise averaging of a mixed quadratic product, so the phase dependence of the rectified response is completely carried by (c, d).

5.2 Two independent mixed kernels

Let $\Theta^{(s)}, \Theta^{(a)}$ denote the unit-amplitude first-order thermal modes and $\widehat{U}^{(s)}, \widehat{V}^{(s)}, \widehat{U}^{(a)}, \widehat{V}^{(a)}$ the corresponding hydrodynamic modes. Defining

$$Z_1 = \widehat{V}^{(a)}\big(\partial_Y \Theta^{(s)}\big)^* + \widehat{U}^{(a)}\big(-ik\Theta^{(s)}\big)^*, \qquad Z_2 = \widehat{V}^{(s)}\big(\partial_Y \Theta^{(a)}\big)^* + \widehat{U}^{(s)}\big(-ik\Theta^{(a)}\big)^*, \tag{52}$$

the streamwise-averaged mixed forcing is exactly

$$\mathcal{E}_{as}(Y \mid A_s, A_a) = c\,\mathcal{K}_c(Y) + d\,\mathcal{K}_q(Y), \qquad \mathcal{K}_c = \tfrac{1}{2}\Re(Z_1 + Z_2), \quad \mathcal{K}_q = \tfrac{1}{2}\,\mathrm{Im}(Z_2 - Z_1). \tag{53}$$

The parity argument of Theorem 1 applies to *each* kernel separately: both $\mathcal{K}_c$ and $\mathcal{K}_q$ are odd in Y, each therefore produces an odd mean temperature correction and wall-flux corrections that are equal and opposite. The in-phase kernel $\mathcal{K}_c$ is the one already used in the published asymmetric configuration, for which $d = 0$ and $\mathcal{E}_{as} = c\,\mathcal{K}_c$ coincides with the form quoted in Section 3. Applying the Green functional of Eq. (41) to the two kernels

defines two independent pumping coefficients,

$$\Pi_T = -\frac{Pe}{2}\int_{-1}^{1} Y\,\mathcal{K}_c(Y)\,dY, \qquad \Pi_T^{\perp} = -\frac{Pe}{2}\int_{-1}^{1} Y\,\mathcal{K}_q(Y)\,dY. \tag{54}$$

Proposition 2 (Exact two-coefficient phase law)**.** *Under the assumptions of Theorem 1, the second-order mean wall-flux redistribution generated by the forcing* (49) *satisfies, exactly at* $O(\varepsilon^2)$*,*

$$\boxed{Nu_{2,0}(+1) = c\,\Pi_T(k, Pe, Gr; Re, Pr) + d\,\Pi_T^{\perp}(k, Pe, Gr; Re, Pr), \qquad Nu_{2,0}(-1) = -Nu_{2,0}(+1).} \tag{55}$$

Both coefficients are linear in Gr *at fixed* (k, Re, Pr)*, and neither admits a higher harmonic in* ϕ*.*

Proof. Eq. (53) expresses the mixed forcing as a linear combination of two fixed profiles with ϕ-independent coefficients c, d; the mean thermal problem and the wall-flux functionals are linear, so the two coefficients are pure numbers depending on (k, Pe, Gr, Re, Pr) only. Odd parity of each kernel gives $Nu_{2,0}(-1) = -Nu_{2,0}(+1)$ as in Theorem 1, and Gr-linearity follows as in Proposition 1 because both kernels are built from the same Gr-linear hydrodynamic modes. Since c and d are, respectively, the real and imaginary parts of a single bilinear product of first-order amplitudes, no other ϕ-dependence can appear at this order. □

Corollary 2 (Pure sinusoidal phase dependence)**.** *For equal wall amplitudes* $\varepsilon_1 = \varepsilon_2 = \varepsilon$*,*

$$c = 0, \qquad d = \tfrac{\varepsilon^2}{2}\sin\phi, \qquad \textit{hence} \qquad Nu_{2,0}(+1\,|\,\phi) = Nu_{2,0}(+1\,|\,\tfrac{\pi}{2})\,\sin\phi. \tag{56}$$

The pumping vanishes at $\phi = 0$ *(purely symmetric) and at* $\phi = \pi$ *(purely antisymmetric), is maximal in magnitude at* $\phi = \pi/2$*, and is odd under* $\phi \to -\phi$*, which is the phase analogue of the amplitude interchange reversal. The proportionality constant is* $\Pi_T^{\perp}$*, not the in-phase coefficient* Π_T*.*

Proof. With $\varepsilon_1 = \varepsilon_2 = \varepsilon$, Eq. (50) gives $A_s = \varepsilon\cos(\phi/2)e^{-i\phi/2}$ and $A_a = i\varepsilon\sin(\phi/2)e^{-i\phi/2}$, whence $A_a A_s^* = i(\varepsilon^2/2)\sin\phi$ is purely imaginary. □

5.3 Numerical values and internal validation

At the reference configuration $(k, Pe, Re, Pr, Gr) = (0.8, 7, 10, 0.7, 100)$ the two coefficients are

$$\Pi_T = -4.74850846\times10^{-2}, \qquad \Pi_T^{\perp} = +8.89496095\times10^{-2}, \qquad \Pi_T^{\perp}/\Pi_T = -1.8732, \tag{57}$$

so the quadrature channel not only differs from the in-phase one, it drives the redistribution in the *opposite* direction and with 1.87 times the magnitude at matched bilinear amplitude. The in-phase value reproduces the authoritative single-case calculation of Section 5 to a relative discrepancy of 4.3×10^{-13}, which validates the kernel decomposition (53) against the published solver.

The law (55) was checked by reconstructing the mixed forcing profile $c\mathcal{K}_c + d\mathcal{K}_q$ and solving the mean thermal problem for twelve randomised triples $(\varepsilon_1, \varepsilon_2, \phi)$; the maximum relative deviation between the algebraic prediction and the direct reconstruction is 5.0×10^{-12}. The sinusoidal form (56) was tested independently of the algebra: the mixed forcing was re-assembled and the mean thermal boundary-value problem re-solved at each of 25 phases spanning $\phi \in [0, \pi]$, and the normalized wall-flux redistribution then lies on $\sin\phi$ to within 2.4×10^{-13}, the in-phase invariant $\Re(A_a A_s^*)$ vanishing to 5.6×10^{-17}. The wall-flux antisymmetry holds for the in-phase and quadrature channels to 6.9×10^{-18} and 1.4×10^{-17} respectively. Both coefficients are linear in Gr: over $Gr = 25$–400 the normalized responses Π_T/Gr and $\Pi_T^{\perp}/Gr$ vary by 1.0×10^{-13} and 3.7×10^{-13} relative respectively, confirming that Corollary 1 covers the new coefficient as well; Sec. 8.1 repeats the scaling in DNS, where the measured Grashof ratio of $q_+ - q_-$ stays within 3.7×10^{-4} of the exact value.

Two structural differences between the channels are worth recording. First, $\Pi_T^{\perp} > 0$ for every $Re \in [5, 50]$ examined at $k = 0.8$ and $Pe = 7$, decreasing monotonically from 1.86×10^{-1} to 8.40×10^{-3}: the quadrature branch shows no reversal in this range, whereas the in-phase branch reverses at $Re_c = 30.838$. The two reversal surfaces in (k, Re) space are therefore distinct. Second, the ratio $\Pi_T^{\perp}/\Pi_T$ is not constant: it varies from -1.17 at $(k, Re) = (0.4, 5)$ to -36.3 at $(1.6, 10)$, attains $+11.0$ at $(0.8, 40)$ and changes sign across the parameter domain, so $\Pi_T^{\perp}$ cannot be reduced to Π_T by any rescaling.

Figure 3 shows the exact phase selection and Figure 4 the two reversal branches. The prediction (56) is a second-order statement and is therefore directly testable in fully nonlinear DNS. With $\varepsilon = 0.03$ and $\phi = \pi/2$ the theorem gives $Nu_{2,0}(+1) = (\varepsilon^2/2)\Pi_T^{\perp} = +4.003 \times 10^{-5}$, whereas the amplitude-asymmetric DNS configuration already validated in Section 6 ($\varepsilon_s = 0.03$, $\varepsilon_a = 0.005$, $\phi = 0$) gives $Nu_{2,0}(+1) = \varepsilon_s \varepsilon_a \Pi_T = -7.123 \times 10^{-6}$. The phase-shifted configuration is thus predicted to pump in the *opposite* direction with 5.6 times the magnitude at the same (k, Pe, Re, Pr, Gr) and the same nominal modulation amplitude. This is an explicit consequence of the theorem, and it has been tested: the DNS runs of Section 9.8 measure $Nu_{2,0}(+1) = +4.0580 \times 10^{-5}$ at $\phi = \pi/2$ and -4.0580×10^{-5} at $\phi = -\pi/2$, 1.38% from the predicted magnitude and antisymmetric to 9.9×10^{-7} of it.

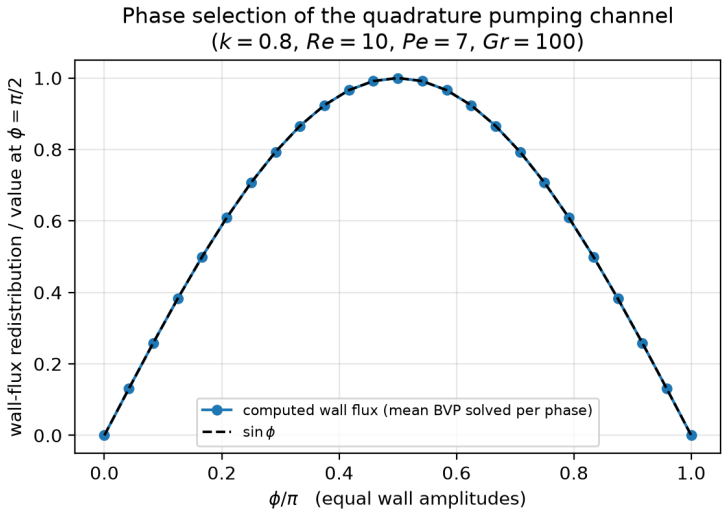


Figure 3: Exact phase selection for equal wall amplitudes at $(k, Pe, Re, Pr, Gr) = (0.8, 7, 10, 0.7, 100)$. Each marker is an independent calculation: the mixed forcing is re-assembled for that phase and the second-order mean thermal problem is re-solved, so the plot is not the analytic expression against itself. The computed normalized response lies on $\sin\phi$ (dashed) to within 2.4×10^{-13}; the response vanishes at the purely symmetric ($\phi = 0$) and purely antisymmetric ($\phi = \pi$) phase relations.

6 Numerical Results and Validation

6.1 Numerical implementation and independent validation

All first-order boundary-value problems are solved using the adaptive collocation algorithm `solve_bvp` from SciPy Virtanen et al. (2020). The mixed second-order temperature field is obtained from Eq. (34). The numerical audit is structured around independent consequences of the analysis: BVP residuals and boundary conditions, reflection parity, forcing parity and integral identities, wall-flux antisymmetry, the Green-function identity, collocation–shooting agreement, mesh sensitivity, and the fixed-bulk-flow condition for the mean streaming correction.

At $(k, Re, Pr, Gr) = (2, 10, 0.7, 100)$, the CPU reference calculation gives

$$Nu_{2,0}(+1) = 1.9730398325 \times 10^{-3}, \qquad Nu_{2,0}(-1) = -1.9730398325 \times 10^{-3}, \tag{58}$$

and hence $\Pi_T = 1.9730398325 \times 10^{-3}$. The independent integral representation agrees to within 3.2×10^{-13}, and the quadrature cross-check is $O(10^{-11})$. The maximum parity residual is below 7×10^{-17} and the wall-flux antisymmetry residual below 9×10^{-19}. Independent collocation–shooting field differences remain below 9×10^{-12}. The fixed-bulk-flow mean streaming correction satisfies its integral constraint to 1.1×10^{-16}.

The reference result is also insensitive to the initial collocation mesh. Calculations with $N_0 = 101, 201, 401, 801$ and a finer reference initialization differ from the reported

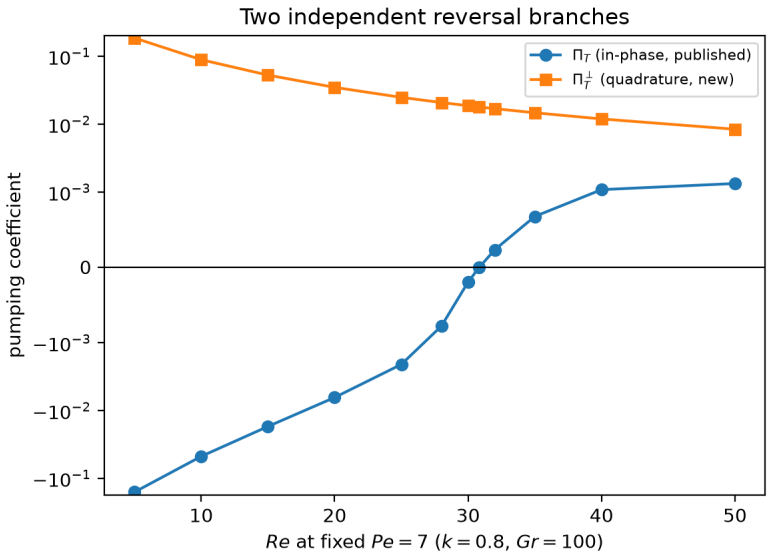


Figure 4: The two independent pumping coefficients versus Reynolds number at fixed $Pe = 7$, $k = 0.8$ and $Gr = 100$. The in-phase coefficient Π_T reverses at $Re_c = 30.838$, while the quadrature coefficient $\Pi_T^{\perp}$ remains positive throughout the scanned interval and decays monotonically.

value only in the last displayed digits. These tests support the use of the adaptive collocation result as the authoritative CPU reference while the shooting calculation supplies an independent field reconstruction. Table 3 collects the diagnostics of that audit.

Table 3: Representative numerical validation diagnostics for the reference case $(k, Re, Pr, Gr) = (2, 10, 0.7, 100)$.

Diagnostic	Maximum reported magnitude
First-order residuals	1.5×10^{-14}
Thermal/velocity parity residuals	6.8×10^{-17}
$\int \mathcal{E}_{as}\, dY$	4.3×10^{-19}
Wall-flux antisymmetry	8.7×10^{-19}
Wall-vs-integral Π_T discrepancy	3.2×10^{-13}
Quadrature cross-check	1.8×10^{-11}
Collocation–shooting difference	8.6×10^{-12}
Zero-bulk-flux residual	1.1×10^{-16}

6.2 Grashof-number scaling

A dedicated sweep at $(k, Re, Pr) = (0.8, 10, 0.7)$ gives a free-intercept fit

$$\Pi_T = (-4.7485 \times 10^{-4})\, Gr - 5.51 \times 10^{-16}, \tag{59}$$

with $R^2 = 1.000000$. The $Gr = 0$ calculation gives zero pumping to numerical precision, confirming Eq. (42); the sweep is plotted in Fig. 5.

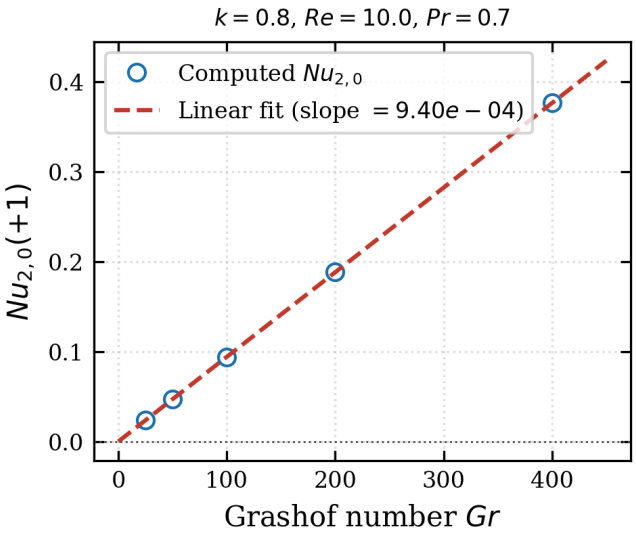


Figure 5: Thermal pumping coefficient versus Grashof number at fixed $(k, Re, Pr) = (0.8, 10, 0.7)$. The free-intercept fit is effectively exact and the response vanishes in the forced-convection limit $Gr = 0$.

6.3 Finite-domain endpoint of the zero-crossing branch

At fixed $Re = 10$ and $Gr = 100$ the normalized response $A = \Pi_T/Gr$ changes sign over the (k, Pe) domain, and the zero set

$$\Pi_T(k^*, Pe) = 0 \tag{60}$$

defines an interior pumping-reversal branch wherever a sign change is present within the admissible interval $0.02 \le k \le 10$. This branch terminates because the interval is finite, not because of a bifurcation, and the endpoint is therefore a property of the computational window.

Two calculations bracket that termination. Just below the endpoint the zero is interior: at $Pe = 10.85$ the response changes sign once, at

$$k^* = 9.338218553141, \qquad |\Pi_T(k^*)| = 1.92 \times 10^{-20}. \tag{61}$$

At the endpoint itself the zero has reached the upper boundary,

$$\boxed{Pe_c = 10.871940554611, \qquad k_c = 10,} \tag{62}$$

where the boundary value $\Pi_T(10, Pe_c) = -8.40 \times 10^{-15}$ is zero to the resolution of the calculation. Just above the endpoint the positive branch is gone: at $Pe = 10.9$ every sampled wavenumber gives a negative response and no sign-change bracket exists, the largest (least negative) value being $\Pi_T(10, 10.9) = -1.4068443 \times 10^{-6}$.

Figure 6 shows the three regimes in one view: interior zero, boundary contact, and

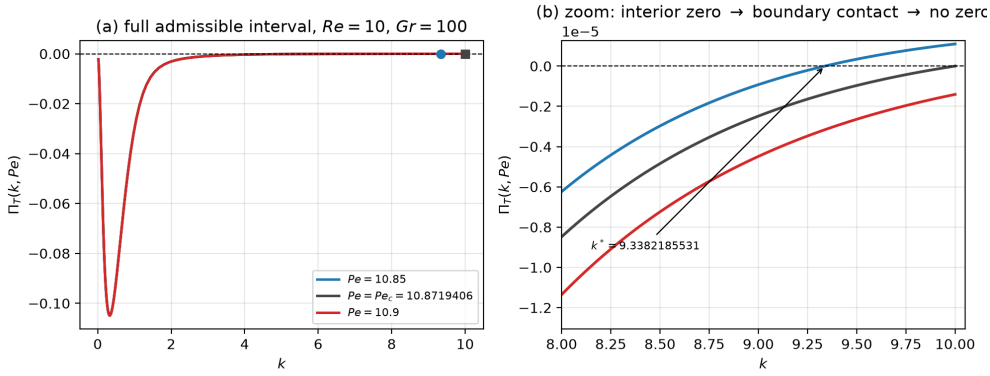


Figure 6: Termination of the interior zero-crossing branch at the finite upper wavenumber boundary ($Re = 10,\ Gr = 100$). (a) Response $\Pi_T(k, Pe)$ over the full admissible interval for the three retained calculations; the curves are visually indistinguishable at this scale, which is itself the point: the change of regime is a near-zero effect confined to $k \gtrsim 8$. (b) High-wavenumber zoom. The interior zero at $k^* = 9.3382185531$ for $Pe = 10.85$ migrates to the boundary $k = 10$ at $Pe = Pe_c$ and disappears for $Pe = 10.9$.

complete loss of positive response inside the window. The endpoint is reached geometrically, when the zero set of $\Pi_T(k, Pe)$ first touches $k = k_{\max} = 10$; it is not prescribed as an independent critical parameter. A dense local check at that boundary confirms that the transition is transversal rather than a sampling artefact: with $\delta = Pe - Pe_c$, the boundary value is $\Pi_T(10, Pe_c - \delta) = +5.00619 \times 10^{-8}$ and $+5.00641 \times 10^{-9}$ for $\delta = 10^{-3}$ and 10^{-4}, and $\Pi_T(10, Pe_c + \delta) = -5.00648 \times 10^{-9}$ and -5.00671×10^{-8} for the same two offsets. The magnitudes are nearly antisymmetric about Pe_c, so the boundary value crosses zero with a finite local slope $\partial\Pi_T/\partial Pe \approx -5.01 \times 10^{-5}$ at $(k, Pe) = (10, Pe_c)$; the full five-row ladder is in supplementary section S1.

The endpoint must therefore be read strictly as a *finite-domain dynamical endpoint* of the observed zero-crossing branch for $(Re, Gr) = (10, 100)$ on $0.02 \le k \le 10$ — not a universal critical Péclet number, not an instability threshold. The survey supporting this classification (173 wavenumbers per Pe for five Pe values, 865 first-order BVP evaluations, plus the local ΔPe ladder quoted above) is archived as supplementary section S1, which records the raw grid, the refined roots, the ΔPe ladder and the run log of that survey; the two bracketing calculations above are the ones used in the main text.

6.4 Mean profile structure

The second-order mean temperature correction and mean streaming velocity provide a complementary picture of the redistribution mechanism. For one-sided asymmetric forcing ($\varepsilon_2 = 0$), the mixed mean thermal correction is odd while the corresponding mean velocity correction reflects the accompanying momentum redistribution under fixed bulk flow.

7 Independent Nonlinear DNS Validation

The perturbation theory developed above is based on the assumption that the wall-temperature modulation amplitude is sufficiently small for the asymptotic expansion to remain valid through second order. Although the analytical results are internally verified by the boundary-value and shooting formulations, an independent nonlinear calculation is required to determine whether the predicted pumping response persists in the fully coupled finite-amplitude transport problem. We therefore perform a separate two-dimensional direct numerical simulation (DNS) of the nonlinear Boussinesq equations.

The DNS does not use the first-order thermal or hydrodynamic modal solutions as input. Instead, the fully nonlinear temperature and momentum equations are integrated directly to a stationary state under the same periodic wall forcing, with the wall heat-flux difference evaluated from the converged temperature field. The DNS is consequently an independent test of the perturbative mechanism rather than a second implementation of the same boundary-value calculation. Four numbers from that campaign carry the argument of Sec. 9.8: the parity cancellation measured at 3.1×10^{-18} in the wall-flux difference, the eight bracketed runs that all return the predicted reversal sign, the Grashof ratio exact to 3.7×10^{-4}, and the quadrature coefficient recovered to 1.38%.

Unless otherwise stated, the reference DNS configuration is

$$(k, Pe, Re, Pr, Gr) = (0.8, 7, 10, 0.7, 100), \tag{63}$$

with

$$\varepsilon_s = 0.03, \qquad \varepsilon_a = 0.005, \tag{64}$$

so that

$$\varepsilon_1 = 0.035, \qquad \varepsilon_2 = 0.025. \tag{65}$$

The nonlinear DNS is performed independently of the perturbative BVP solver. The stationary state is accepted only after the residual of the discretized evolution equations falls below the prescribed convergence threshold and the wall-flux observables become stationary.

The DNS estimator mirrors the perturbative definition (39) and is formed from the cycle-averaged wall-normal wall heat fluxes $q_\pm$ alone. Because $q_+ - q_-$ is the fully nonlinear counterpart of $Nu_{2,0}(+1) - Nu_{2,0}(-1)$, the coefficient recovered from the simulation is

$$\Pi_T^{\mathrm{DNS}} \;=\; \frac{q_+ - q_-}{2\varepsilon_a \varepsilon_s}. \tag{66}$$

For the reference configuration, $q_+ - q_- = -1.40289979 \times 10^{-5}$ and $2\varepsilon_a\varepsilon_s = 3.0 \times 10^{-4}$, giving $\Pi_T^{\mathrm{DNS}} = -4.6763 \times 10^{-2}$, the value quoted in Table 5. No first-order modal field enters the estimator, which is what makes the comparison independent of the perturbative

solver.

7.1 Amplitude convergence and recovery of the perturbative limit

The first and most direct test concerns the asymptotic limit $\varepsilon_a \to 0$. Because the leading pumping contribution is proportional to the mixed product $\varepsilon_a \varepsilon_s$, the normalized coefficient extracted from the nonlinear DNS should approach a finite limit as the asymmetry amplitude decreases. We therefore performed a sequence of DNS calculations with decreasing ε_a while keeping $\varepsilon_s = 0.03$ fixed.

The resulting values of the nonlinear pumping coefficient are

ε_a	Π_T^{DNS}
0.0150	$-4.67629365 \times 10^{-2}$
0.0100	$-4.67631801 \times 10^{-2}$
0.0075	$-4.67632654 \times 10^{-2}$
0.0050	$-4.67633263 \times 10^{-2}$

(67)

and the extrapolation to $\varepsilon_a \to 0$ gives

$$\Pi_T^{\mathrm{DNS},\,\varepsilon_a \to 0} \simeq -4.6763 \times 10^{-2}. \tag{68}$$

The perturbative prediction for the same reference state is

$$\Pi_T^{(2)} = -4.7485 \times 10^{-2}, \tag{69}$$

corresponding to a relative difference of approximately 1.52%.

This agreement is significant for two reasons. First, the DNS solves the nonlinear problem at finite wall-modulation amplitude rather than the asymptotically truncated equations. Second, the observed convergence of Π_T^{DNS} as ε_a decreases demonstrates that the nonlinear solution approaches a finite coefficient consistent with the second-order theory. The result therefore supports the interpretation of Π_T as the leading nonlinear transport coefficient associated with the symmetric–antisymmetric interaction (Fig. 7).

7.2 Spatial and temporal convergence

The numerical reliability of the DNS observable is assessed independently through grid and time-step refinement. For the spatial study, the calculation was performed on progressively refined meshes 48×37, 64×49, 80×61, and 96×73. The corresponding pumping

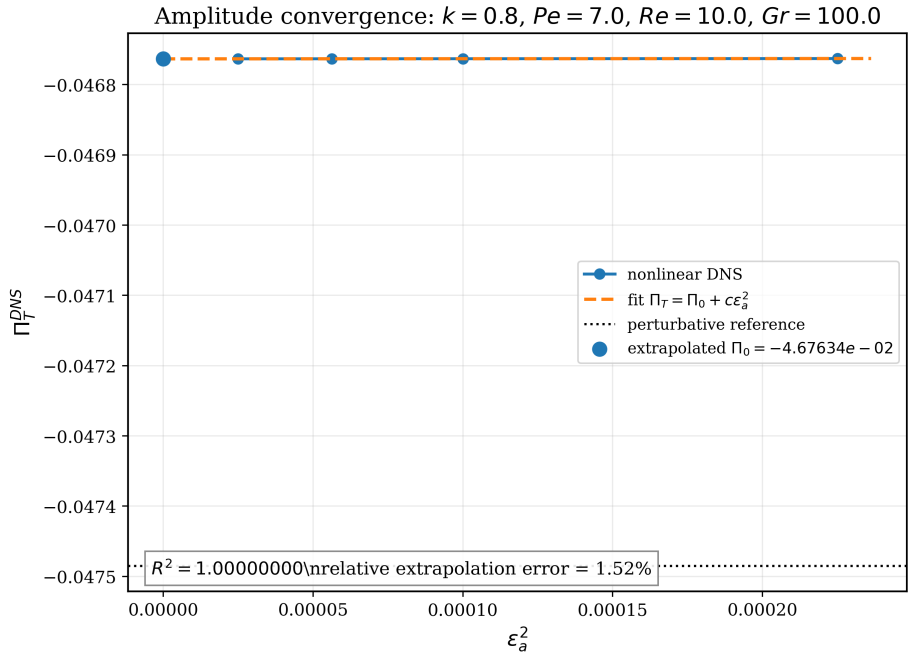


Figure 7: Convergence of the fully nonlinear DNS pumping coefficient with decreasing asymmetry amplitude. The nonlinear DNS values approach a finite $\varepsilon_a \to 0$ limit, which differs by only 1.52% from the second-order perturbative prediction. The result provides an independent nonlinear validation of the asymptotic pumping coefficient.

coefficients are

$N_x \times N_y$	Π_T^{DNS}	relative difference from finest grid
48×37	$-4.65557743 \times 10^{-2}$	1.10%
64×49	$-4.67633263 \times 10^{-2}$	0.66%
80×61	$-4.69453055 \times 10^{-2}$	0.27%
96×73	$-4.70739712 \times 10^{-2}$	0

(70)

where the finest mesh is used only as the numerical reference for the convergence study. The monotonic variation toward the finest-grid solution indicates that the wall-flux observable is spatially resolved rather than being controlled by under-resolution of the transverse boundary layers (Fig. 8).

The temporal convergence study gives

Δt	Π_T^{DNS}	relative difference from smallest Δt
3.0×10^{-3}	$-4.67632768 \times 10^{-2}$	1.75×10^{-6}
2.0×10^{-3}	$-4.67633588 \times 10^{-2}$	1.38×10^{-9}
1.5×10^{-3}	$-4.67633263 \times 10^{-2}$	6.95×10^{-7}
1.0×10^{-3}	$-4.67633588 \times 10^{-2}$	0

(71)

The extremely small change in Π_T over this sequence shows that the reported pumping

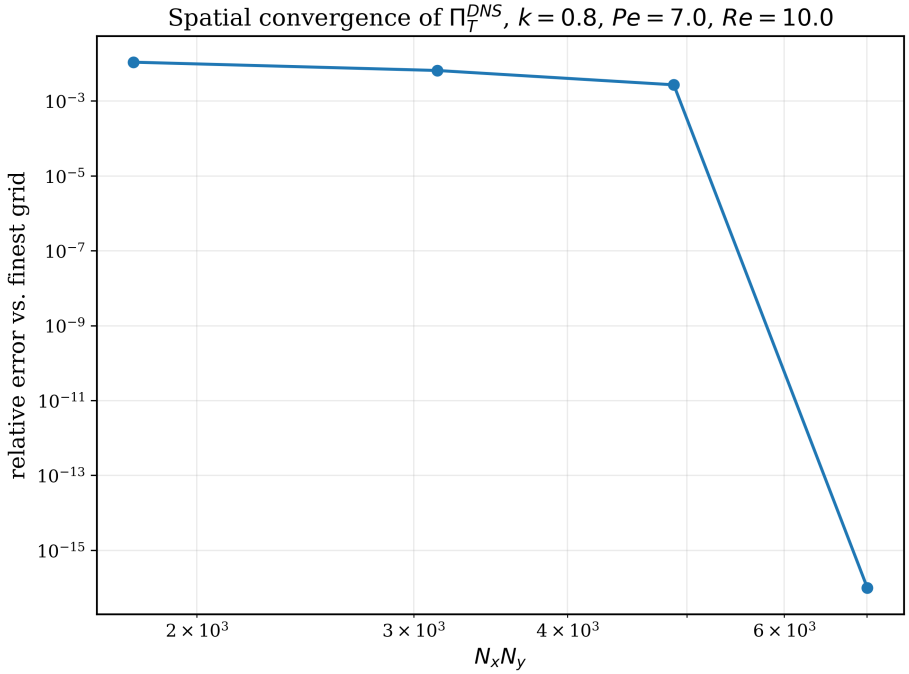


Figure 8: Spatial convergence of the nonlinear DNS pumping coefficient. The solution approaches a well-defined finest-grid value as both the streamwise and transverse resolutions are increased. The remaining difference between the finest-grid DNS result and the perturbative reference is smaller than the total discrepancy observed on coarser meshes.

coefficient is not controlled by the chosen temporal discretization (Fig. 9).

The convergence histories provide an additional stationarity check. At the reference resolution, the final evolution residual is

$$\mathcal{R}_{\mathrm{DNS}} \lesssim 7 \times 10^{-9}, \tag{72}$$

while the global wall-flux balance satisfies

$$q_+ + q_- \simeq -1.9 \times 10^{-7}. \tag{73}$$

These values show that the reported Π_T is extracted from a genuinely converged stationary state rather than from an arbitrary finite-time snapshot.

7.3 Independent verification of the Grashof-number scaling

The analytical formulation predicts that the first-order hydrodynamic response is proportional to Gr and therefore that, at fixed (k, Re, Pr),

$$\Pi_T = GrA(k, Re, Pr). \tag{74}$$

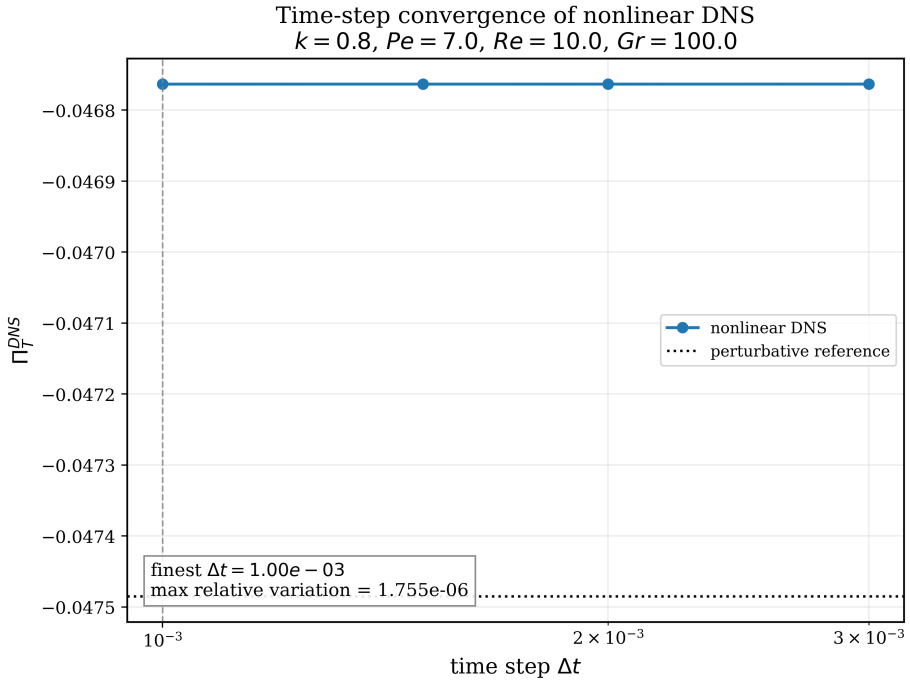


Figure 9: Time-step convergence of the nonlinear DNS pumping coefficient. The response remains unchanged to within approximately 10^{-6} in relative terms over the tested time-step range, confirming that the stationary wall-flux observable is temporally converged.

This prediction is tested directly with fully nonlinear DNS by varying Gr from zero to 200 while retaining the same geometry, Reynolds number, and thermal parameters.

The DNS gives a practically linear dependence over the investigated range, with a coefficient of determination indistinguishable from unity at the reported numerical precision. In addition, the zero-buoyancy simulation gives

$$\Pi_T(Gr = 0) \simeq -1.1 \times 10^{-14}, \tag{75}$$

which is numerically zero (Fig. 10).

This result is particularly useful because it validates a structural prediction that is independent of the spatial-parity proof. The symmetry theorem determines the admissible relation between the two wall-flux corrections, while the present DNS confirms independently that the strength of this redistribution is controlled linearly by the buoyancy amplitude at leading order.

7.4 Reynolds-number control at fixed Péclet number

A central consequence of the governing equations is that Pe cannot in general be treated as the sole control parameter for the pumping coefficient. The thermal modal problem depends on $Pe = Re\,Pr$, but the hydrodynamic response retains explicit dependence on

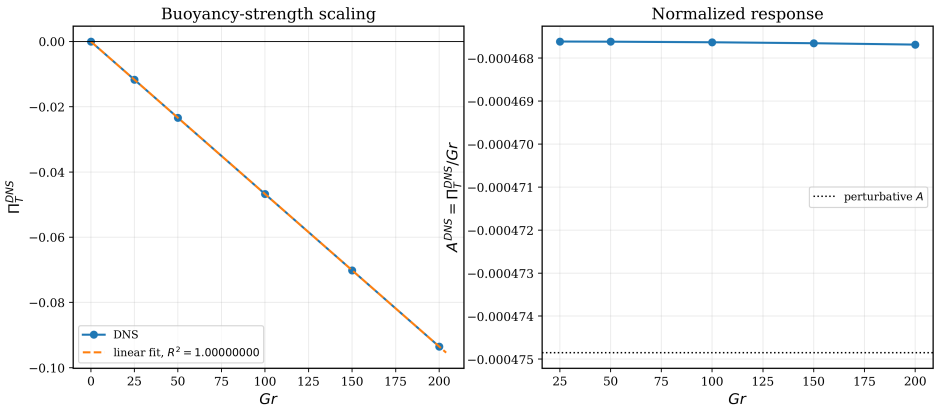


Figure 10: Nonlinear DNS verification of the Grashof-number scaling at fixed $(k, Re, Pr) = (0.8, 10, 0.7)$. The pumping coefficient is linear in Gr over the investigated interval, while the $Gr = 0$ calculation switches off the leading pumping response to numerical precision. The result independently confirms the analytical proportionality $\Pi_T = GrA(k, Re, Pr)$.

Re. The fully nonlinear DNS provides an independent physical test of this distinction.

We therefore keep

$$Pe = 7, \qquad Gr = 100, \qquad k = 0.8 \tag{76}$$

fixed and vary Re, adjusting the Prandtl number according to

$$Pr = \frac{Pe}{Re}. \tag{77}$$

The normalized pumping response obtained from DNS is

Re	Pr	$A_{\mathrm{DNS}} = \Pi_T/Gr$	$\mathrm{sign}(A_{\mathrm{DNS}})$
5	1.4000	-1.5878×10^{-3}	$-$
10	0.7000	-4.6763×10^{-4}	$-$
20	0.3500	-5.9977×10^{-5}	$-$
30	0.2333	$+7.5042 \times 10^{-7}$	$+$
35	0.2000	$+8.9842 \times 10^{-6}$	$+$

(78)

The independent DNS therefore reproduces a clear reversal of the pumping direction. The response is negative at $Re = 20$ but has become positive by $Re = 30$, demonstrating directly that changing the hydrodynamic Reynolds number while preserving the same Pe can reverse the direction of nonlinear thermal pumping (Fig. 11).

The corresponding perturbative values of A at $Re = 5$, 10, 20, and 35 are -1.5951×10^{-3}, -4.7485×10^{-4}, -6.4279×10^{-5}, and $+6.7452 \times 10^{-6}$, respectively. The quantitative agreement is strongest at the lower and intermediate Reynolds numbers, with relative differences of 0.46%, 1.52%, and 6.69% for $Re = 5$, 10, and 20. At $Re = 35$, the relative difference increases to 33.19%; however, both the DNS and perturbative responses are

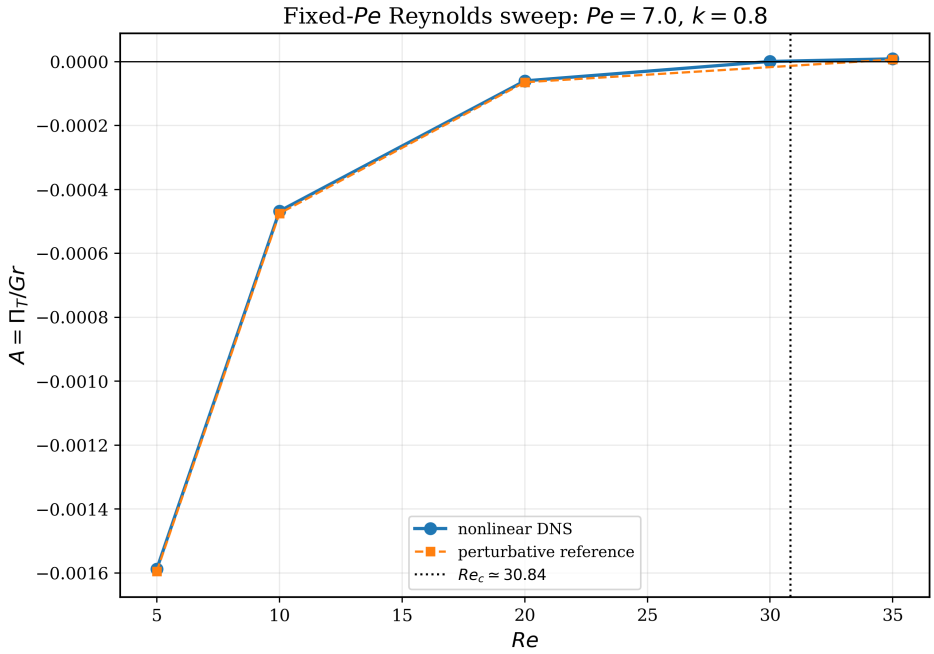


Figure 11: Normalized pumping response $A = \Pi_T/Gr$ from the independent nonlinear DNS at fixed $Pe = 7$, $k = 0.8$, and $Gr = 100$. Varying Re changes both the magnitude and the sign of the response. The transition from negative values at $Re = 20$ to positive values at $Re = 30$ and 35 provides direct nonlinear evidence that the hydrodynamic Reynolds number remains an active control parameter even when the Péclet number is held fixed.

very small in absolute magnitude and have the same positive sign. The discrepancy therefore reflects the sensitivity of a relative error metric near a zero crossing rather than a disagreement in the qualitative pumping mechanism. Near the reversal, the sign of the response is therefore a more robust diagnostic than a relative error normalized by a quantity approaching zero.

The principal physical result of this sweep is thus not the absolute value at $Re = 35$, but the independently confirmed reversal of the pumping direction. This observation supports the analytical conclusion that A is, in general, a function of (k, Re, Pr) and cannot be reduced universally to a function of (k, Pe) alone; Table 4 lists the sweep.

Table 4: Independent nonlinear DNS at fixed $Pe = 7$, $k = 0.8$, and $Gr = 100$. The Prandtl number is adjusted according to $Pr = Pe/Re$.

Re	Pr	A_{DNS}	A_{pert}	Relative difference
5	1.4000	-1.5878×10^{-3}	-1.5951×10^{-3}	0.46%
10	0.7000	-4.6763×10^{-4}	-4.7485×10^{-4}	1.52%
20	0.3500	-5.9977×10^{-5}	-6.4279×10^{-5}	6.69%
30	0.2333	$+7.5042 \times 10^{-7}$	–	–
35	0.2000	$+8.9842 \times 10^{-6}$	$+6.7452 \times 10^{-6}$	33.19%

7.5 Independent verification of reflection symmetry and conservation

The exact parity theorem predicts two particularly strong properties that can be tested without comparing the DNS with any perturbative amplitude: (i) reflection of the upper and lower wall amplitudes must reverse the direction of the wall-flux redistribution, and (ii) perfectly symmetric wall forcing must suppress the directional pumping observable.

For the asymmetric reference case, the DNS gives

$$q_+ - q_- = -1.40289979 \times 10^{-5}, \tag{79}$$

whereas after interchanging the upper and lower wall amplitudes,

$$q_+ - q_- = +1.40289979 \times 10^{-5}. \tag{80}$$

The corresponding reflection residual is

$$|D_{\text{reflected}} + D_{\text{baseline}}| = 3.18 \times 10^{-17}. \tag{81}$$

For perfectly symmetric wall forcing, the DNS gives

$$q_+ - q_- = 8.13 \times 10^{-18}, \tag{82}$$

which is indistinguishable from zero at the resolution of the simulation (Fig. 12).

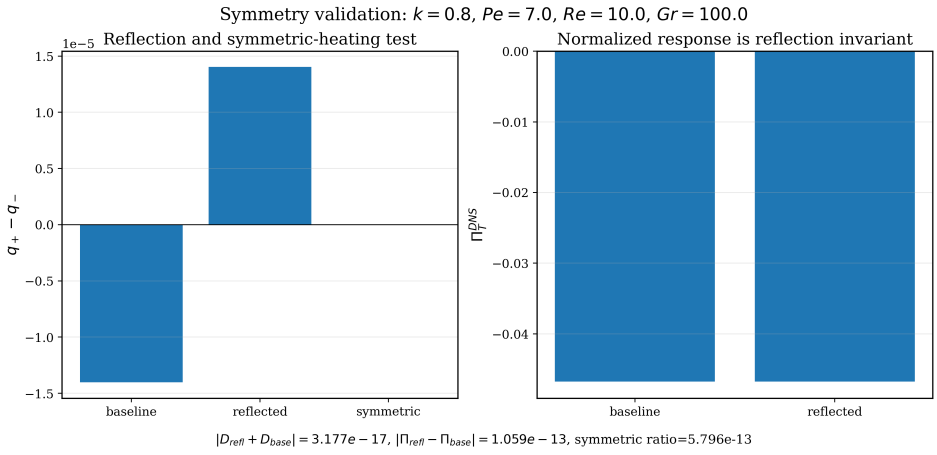


Figure 12: Independent DNS verification of reflection symmetry and wall-flux conservation. Interchanging the upper and lower thermal forcing reverses the sign of the wall-flux difference, whereas perfectly symmetric forcing suppresses the directional redistribution. The measured symmetry residuals are at or below 10^{-13}, with the direct flux-reflection residual reaching 3.18×10^{-17}.

The DNS therefore reproduces the structural symmetry law without explicitly enforc-

ing the analytical parity relations. This is an important distinction: the perturbative derivation proves the selection rule analytically, whereas The nonlinear DNS independently recovers the same reflection-induced reversal of the wall-flux difference and the suppression of directional redistribution under symmetric forcing. Thus, the symmetry-selected structure identified analytically is also observed in the fully coupled nonlinear calculation at finite modulation amplitude.

7.6 Consolidated validation

Figure 13 summarizes the independent numerical evidence. The combination of amplitude extrapolation, spatial refinement, time-step refinement, Grashof-number scaling, Reynolds-number sweeps, reflection tests, and wall-flux conservation provides validation at three complementary levels.

First, the amplitude and resolution studies establish numerical convergence of the nonlinear observable. Second, the Gr and symmetry studies verify the structural consequences of the analytical framework. Third, the fixed-Pe Reynolds sweep provides a genuinely physical result that is not contained in the parity proof alone: the direction of nonlinear pumping can be selected by the hydrodynamic Reynolds number while the thermal Péclet number is unchanged.

For convenience, the principal validation results are collected in Fig. 13 and Table 5. The spatial and temporal entries are reported as differences from the finest calculation in the corresponding convergence sequence; they therefore do not represent formal discretization error bounds.

Table 5: Summary of the independent nonlinear DNS validation campaign. Spatial and temporal convergence values are measured relative to the finest calculation in each sequence. Reflection and symmetric-forcing residuals are absolute numerical residuals.

Test	DNS result	Reference or criterion	Assessment
Amplitude extrapolation	-4.6763×10^{-2}	-4.7485×10^{-2}	1.52% difference
Spatial convergence	$< 1.10\%$	finest-grid limit	converged
Time-step convergence	$< 1.8 \times 10^{-6}$	smallest-Δt limit	converged
Gr scaling	$R^2 \simeq 1$	linearity	confirmed
$Gr = 0$	$\lvert\Pi_T\rvert \simeq 1.1 \times 10^{-14}$	0	confirmed
Reflection flux residual	3.18×10^{-17}	0	confirmed
Reflection Π_T residual	1.06×10^{-13}	0	confirmed
Symmetric forcing residual	5.80×10^{-13}	0	confirmed

Taken together, these calculations provide an independent nonlinear verification of the main analytical conclusions without requiring the DNS to reproduce every perturbative coefficient to arbitrary relative accuracy near a zero crossing. The strongest quantitative agreement occurs away from the reversal region, whereas the vicinity of the reversal is characterized by excellent agreement in the sign and by enhanced sensitivity of the

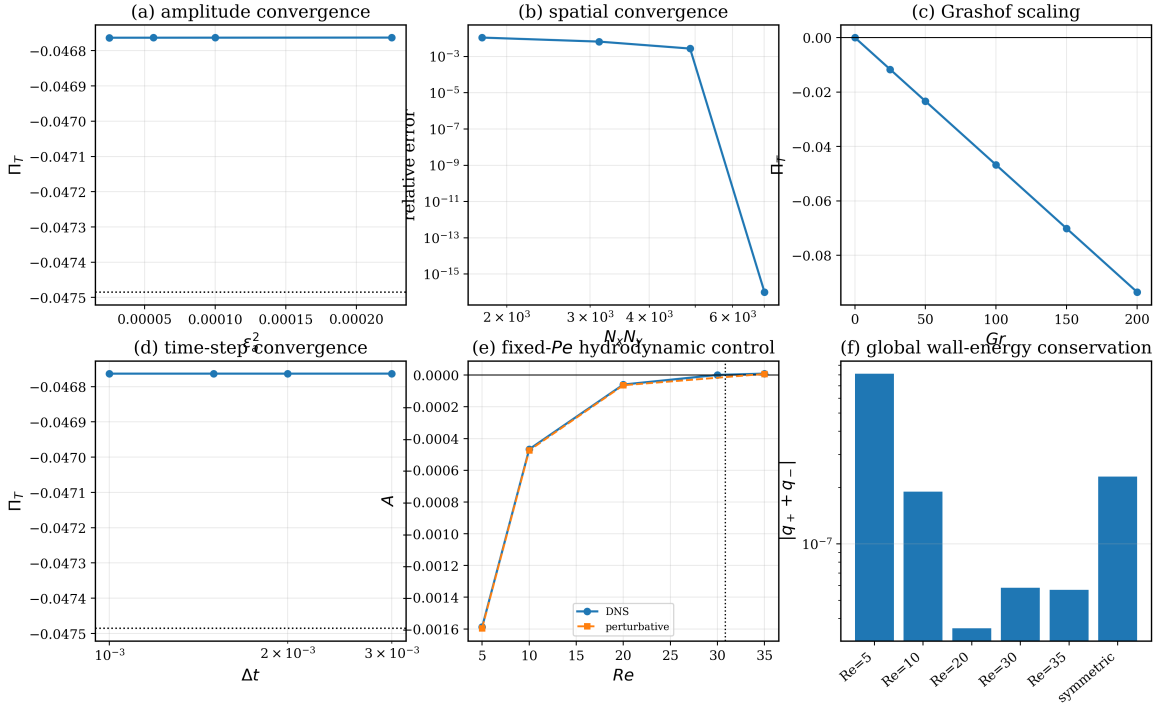


Figure 13: Consolidated summary of the independent nonlinear DNS validation campaign. The seven diagnostic panels combine amplitude extrapolation, spatial and temporal convergence, Grashof-number scaling, fixed-Pe Reynolds dependence, and wall-flux conservation. Together they show that the perturbative thermal-pumping framework is recovered by an independent nonlinear computation and that the Reynolds number remains an active control parameter for the direction of the pumping response.

relative error to small absolute differences. This behavior is physically consistent with the interpretation of the pumping response as the result of competing thermal–hydrodynamic pathways.

7.7 Physical implication of the nonlinear DNS validation

The DNS results strengthen the physical interpretation of the pumping mechanism by demonstrating that the symmetry-selected wall-flux redistribution is not an artifact of the perturbative truncation. The exact reflection properties survive in the fully nonlinear equations, while the magnitude and direction of the response remain dynamically selectable. In particular, the fixed-Pe Reynolds sweep shows that the thermal field and hydrodynamic mediator should not be viewed as a single reduced transport process: Pe controls the thermal competition between advection and diffusion, whereas Re independently modulates the buoyancy-induced velocity field that transports the asymmetric thermal mode. The observed reversal of Π_T therefore provides a physical signature of competing hydrodynamic pathways underlying the symmetry-selected thermal redistribution.

8 Systematic Mapping of the Reynolds-Number Reversal

The reversal reported above is obtained from fully nonlinear DNS at a single wavenumber, and a single point cannot distinguish a law from an accident. We therefore map the reversal locus $Re_c(k)$ with the authoritative perturbative solver over seven wavenumbers at fixed $Pe = 7$ and $Gr = 100$, with $Pr = Pe/Re$, scanning $Re \in \{5, 10, 15, 20, 25, 30, 35, 40\}$ and refining each sign change by Brent iteration. Because the two longest-wavelength crossings fall outside that interval, the scan was extended for $k = 0.4$ and $k = 0.6$ up to $Re = 120$. Section 9.8 confirms this locus nonlinearly: DNS run on both sides of the crossing at four of the seven wavenumbers returns the predicted sign in every case. The complete grid and the seven refined crossings are archived with this paper as supplementary section S2.

Table 6: Reversal Reynolds number at fixed $Pe = 7$ and $Gr = 100$. The five entries with $Re_c \leq 40$ are bracketed inside the primary scan; the two longest-wavelength crossings lie beyond $Re = 40$ and were bracketed on the extended interval. The reversal is therefore resolved at every wavenumber sampled. The $k = 0.8$ value reproduces the independent determination from the DNS-based sweep, $Re_c = 30.838496518$, to nine significant digits.

k	0.4	0.6	0.8	1.0	1.2	1.5	2.0
$Re_c(k)$	79.080214	51.252439	30.838497	20.487029	15.089038	11.069430	8.541316

Two facts follow. First, the reversal is systematic rather than isolated: it occurs at every one of the seven wavenumbers sampled and its location moves monotonically to lower Re as k increases, from $Re_c = 79.08$ at $k = 0.4$ to $Re_c = 8.54$ at $k = 2.0$. Over the seven resolved crossings the empirical description $Re_c \approx 21.50\, k^{-1.47}$ holds with a maximum relative deviation of 11.0%; restricting the fit to the five crossings of the primary $Re \leq 40$ window gives $Re_c \approx 20.91\, k^{-1.41}$ with a maximum deviation of 7.7%, so the two extra points do not follow the same power law as closely as the interior ones. We report both as fits, not as derived scalings, because no exponent has been obtained analytically. Second, the reversal boundary of the (k, Re) plane is not a finite wall at $Re \approx 40$: it recedes toward large Re as k decreases, and at $Gr = 100$ the longest waves reverse only in the low-Pr corner of the (Re, Pr) plane ($Re = 79.08$ and $Pr = 0.089$ at $k = 0.4$). Fig. 14 draws the seven responses whose crossings define that boundary, and Fig. 15 the refined locus with its power-law fit.

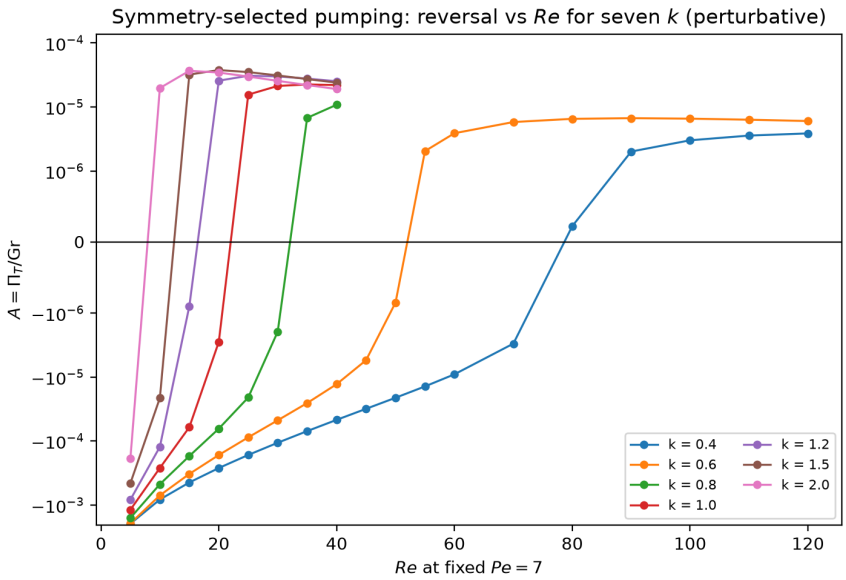


Figure 14: Normalized pumping response $A = \Pi_T/Gr$ versus Reynolds number at fixed $Pe = 7$ and $Gr = 100$ for seven wavenumbers, computed with the authoritative perturbative solver. Each curve crosses zero exactly once. The $k = 0.4$ and $k = 0.6$ curves are drawn over the extended interval $Re \leq 120$ that their crossings require; the remaining five use the primary $Re \leq 40$ window, inside which each of them already crosses. The crossing moves to lower Re as k increases.

8.1 Grashof-independence of the reversal: direct test of Corollary 1

Corollary 1 predicts that the reversal location does not move when the buoyancy amplitude is changed. This is tested at $k = 0.8$, $Pe = 7$ by locating Re_c independently for four Grashof numbers.

Table 7: Test of Corollary 1 at $k = 0.8$, $Pe = 7$. The normalized response is invariant to a worst-case relative spread of 7.3×10^{-11} and the reversal Reynolds number is identical to the nine displayed digits for all four Grashof numbers (absolute spread 1.2×10^{-10}); the scan is archived as supplementary section S2.

Gr	$A(Re = 25)$	$A(Re = 30)$	$A(Re = 35)$	Re_c
25	-2.089882×10^{-5}	-1.958845×10^{-6}	$+6.745165 \times 10^{-6}$	30.838496520
50	-2.089882×10^{-5}	-1.958845×10^{-6}	$+6.745165 \times 10^{-6}$	30.838496520
100	-2.089882×10^{-5}	-1.958845×10^{-6}	$+6.745165 \times 10^{-6}$	30.838496520
200	-2.089882×10^{-5}	-1.958845×10^{-6}	$+6.745165 \times 10^{-6}$	30.838496520

The invariance is exact to the convergence tolerance of the solver, which is the strongest form of confirmation available for a structural prediction: the reversal point is a property of $A(k, Re, Pr)$ and not of the buoyancy amplitude; the four branches are overlaid in Fig. 16. The same test applied to the quadrature coefficient $\Pi_T^{\perp}$ gives an invariant $\Pi_T^{\perp}/Gr$ to 3.7×10^{-13} over $Gr = 25$–400.

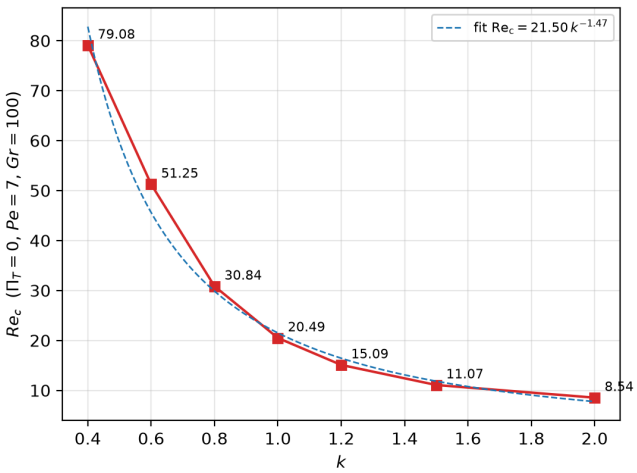


Figure 15: Refined reversal locus $Re_c(k)$ at $Pe = 7$, $Gr = 100$ (Brent iteration on the perturbative solver), resolved at all seven sampled wavenumbers; the dashed line is the seven-point power-law fit $Re_c = 21.50\,k^{-1.47}$. The $k = 0.8$ point coincides with the value previously obtained from the fixed-Pe critical-Reynolds analysis.

The same statement can be tested without the perturbative solver. A 2×2 DNS matrix at $k = 0.8$ and $Pe = 7$ pairs $Gr = 50$ with $Gr = 200$ at two Reynolds numbers straddling Re_c, one below and one above. The measured Grashof ratios of $q_+ - q_-$ are 4.0003 at $Re = 20$ and 4.0015 at $Re = 50$, against the exact 4 required by Corollary 1: a worst-case relative error of 3.7×10^{-4}, while the response amplitude itself changes by a factor of four. The sign is unchanged within each pair—negative at $Re = 20$, positive at $Re = 50$—so the reversal does not move when the buoyancy amplitude is quadrupled in the fully nonlinear regime. The offset between DNS and the perturbative coefficient is itself proportional to Gr (the two deviations of a pair differ by 3.996 and 4.015), which is why it rescales the response without displacing its zero: the relative error of $A = \Pi_T/Gr$ is the same at both Grashof numbers, 6.69% at $Re = 20$ and 10.7% at $Re = 50$. The checkpoint record of the four runs, archived as supplementary section S3, reduces to Table 8 and to Fig. 17, which draws the same statement as a single branch: the reduced coefficient $A = \Pi_T/Gr$ measured at the two Grashof numbers falls on one curve, and that curve crosses zero at the perturbative Re_c. The four runs cost 491 s in total, with residuals below 1.0×10^{-8}.

The design is not accidental: candidate Reynolds numbers closer to the crossing were rejected before integration by the resolvability gate of Sec. 9.9, which requires the predicted signal to exceed the deviation floor by more than a factor of two. The pair $Re = 32$ has margins 0.43 and 1.72 and the pair $Re = 40$ has 1.99 and 7.98, so in both cases at least one member cannot test a sign; the accepted pair keeps margins between 2.5 and 47.4 on both sides of Re_c.

The residual departures of the two measured ratios from exactly four—0.1% and 0.4%—are not interpreted. A Richardson-type correction Gr/Re^2 would predict the larger departure at the smaller Reynolds number, since $Gr/Re^2 = 0.50$ for $(Gr, Re) = (200, 20)$

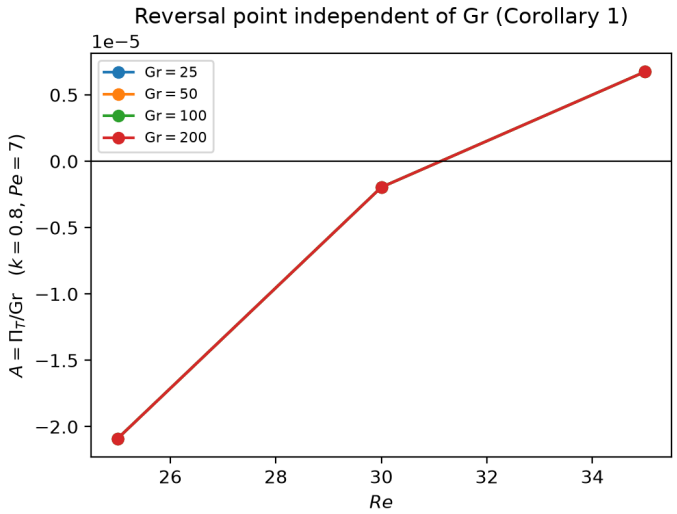


Figure 16: Normalized response $A = \Pi_T/Gr$ versus Re at $k = 0.8$ and $Pe = 7$ for four Grashof numbers. The four curves coincide, as required by Corollary 1.

Table 8: Nonlinear test of Corollary 1 at $k = 0.8$, $Pe = 7$, $\varepsilon_s = 0.030$, $\varepsilon_a = 0.005$ and $\phi = 0$, with $Pr = Pe/Re$. Exact linearity in Gr forces the ratio of the two measured flux differences at each Re to equal $200/50 = 4$; the ratios are 4.0003 and 4.0015. The margin column is $|\Pi_T|$ in units of the deviation model δ_Π; the raw record is archived as supplementary section S3.

Re	Gr	Pr	$q_+ - q_-$ predicted	measured	margin
20	50	0.3500	-9.641824×10^{-7}	-8.996526×10^{-7}	11.9
20	200	0.3500	-3.856730×10^{-6}	-3.598839×10^{-6}	47.4
50	50	0.1400	$+1.994872 \times 10^{-7}$	$+2.207512 \times 10^{-7}$	2.5
50	200	0.1400	$+7.979489 \times 10^{-7}$	$+8.833327 \times 10^{-7}$	9.8

against 0.08 for $(200, 50)$, whereas the observed ordering is the reverse. The ratios are therefore reported as consistent with exact Grashof linearity at the tolerance the runs reach, and the structural claim rests on the proportionality of the *discrepancy* together with the unchanged sign, not on the fourth decimal of the ratio.

9 Physical Interpretation and Discussion

9.1 Quantitative phase analysis of the reversal

The reversal must be explained, not merely reported. Because the mean thermal problem and the wall-flux functional $\mathcal{L}[\cdot]$ are linear, the four terms of the mixed forcing (52) contribute additively to Π_T:

$$\Pi_T = \mathcal{L}\big[\langle V^{(a)}\partial_Y\Theta^{(s)}\rangle\big] + \mathcal{L}\big[\langle U^{(a)}\partial_X\Theta^{(s)}\rangle\big] + \mathcal{L}\big[\langle V^{(s)}\partial_Y\Theta^{(a)}\rangle\big] + \mathcal{L}\big[\langle U^{(s)}\partial_X\Theta^{(a)}\rangle\big], \qquad (83)$$

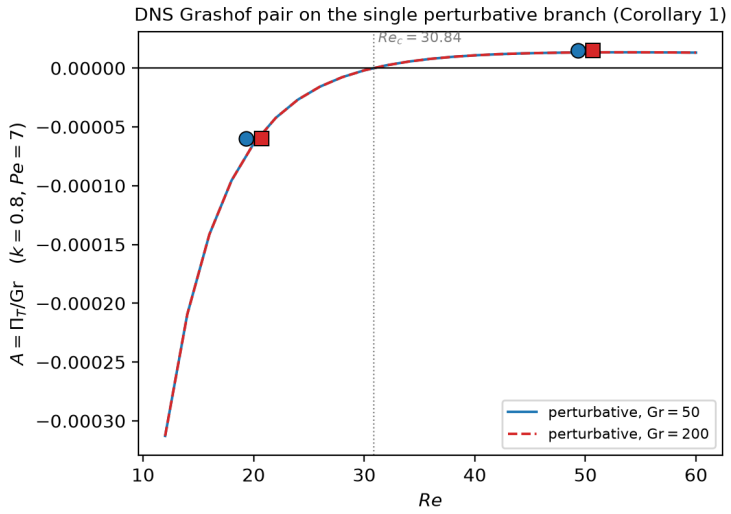


Figure 17: Nonlinear reading of Corollary 1: the reduced coefficient $A = \Pi_T/Gr$ versus Re at $k = 0.8$, $Pe = 7$. The two perturbative branches, $Gr = 50$ (solid) and $Gr = 200$ (dashed), are computed independently and separate by at most 1.1×10^{-10} in relative terms, which is the Gr-independence of A and of its zero $Re_c = 30.84$. The markers are the four DNS measurements of Table 8: circles at $Gr = 50$, squares at $Gr = 200$, offset by ± 0.7 in Re for legibility. Within each pair the two DNS values of A agree to 6.3×10^{-5} ($Re = 20$) and 3.7×10^{-4} ($Re = 50$), so the nonlinear response falls on the same single branch at both buoyancy amplitudes; the branch itself is approached to 6.7% and 10.7%, the deviation analysed in Sec. 9.9. The measured values and the reduced data of the two branches are archived as supplementary section S3.

where $\langle\cdot\rangle$ denotes streamwise averaging and each term is evaluated with the other mode at unit amplitude. Eq. (83) is an identity, not a model: evaluating the four functionals separately reproduces the full Π_T with a maximum absolute deviation of 1.4×10^{-14} over the whole Re scan (the relative deviation reaches 1.6×10^{-5} only at $Re = Re_c$, where Π_T itself vanishes; Table 9).

Three conclusions follow, and one common expectation is disproved.

1. **The reversal is a magnitude race, not a phase flip.** The ratio of the two dominant contributions, $\left|\mathcal{L}[\langle V^{(a)}\partial_Y\Theta^{(s)}\rangle]\right|/\mathcal{L}[\langle U^{(a)}\partial_X\Theta^{(s)}\rangle]$, falls monotonically over the whole scan, from 3.40 at $Re = 5$ through 1.012 at the reversal itself to 0.936 at $Re = 50$. Neither term changes sign anywhere in the scan; only their relative size does.

2. **The subdominant terms are not negligible at the crossing.** The two-term balance alone would predict the reversal at $Re = 32.713$, whereas the full sum crosses at $Re_c = 30.838$; the difference of 1.88 in Re (6.1%) is caused by the two symmetric-velocity terms, whose net contribution is $+2.7 \times 10^{-4}$ at the crossing—about 1.2% of the dominant terms but the entire residual that decides the sign. Any two-term explanation of the reversal is therefore quantitatively wrong.

Table 9: Term-wise balance of Π_T at $k = 0.8$, $Pe = 7$, $Gr = 100$ with $Pr = Pe/Re$. The two dominant contributions keep fixed signs—the streamwise advection term is positive, the wall-normal sweep term negative—and the reversal occurs when the faster decay of the sweep term lets the two cancel; the term-by-term evaluation is archived as supplementary section S2.

Re	$\mathcal{L}[\langle U^{(a)}\partial_X\Theta^{(s)}\rangle]$	$\mathcal{L}[\langle V^{(a)}\partial_Y\Theta^{(s)}\rangle]$	$\mathcal{L}[\langle U^{(s)}\partial_X\Theta^{(a)}\rangle]$	$\mathcal{L}[\langle V^{(s)}\partial_Y\Theta^{(a)}\rangle]$	Π_T
5	5.504×10^{-2}	-1.870×10^{-1}	-2.537×10^{-2}	-2.236×10^{-3}	-1.595×10^{-1}
10	5.899×10^{-2}	-9.726×10^{-2}	-2.707×10^{-3}	-6.504×10^{-3}	-4.749×10^{-2}
20	3.737×10^{-2}	-4.251×10^{-2}	$+4.895 \times 10^{-3}$	-6.180×10^{-3}	-6.428×10^{-3}
30	2.378×10^{-2}	-2.420×10^{-2}	$+4.829 \times 10^{-3}$	-4.611×10^{-3}	-1.959×10^{-4}
Re_c	2.300×10^{-2}	-2.326×10^{-2}	$+4.761 \times 10^{-3}$	-4.494×10^{-3}	≈ 0
40	1.655×10^{-2}	-1.599×10^{-2}	$+3.939 \times 10^{-3}$	-3.420×10^{-3}	$+1.081 \times 10^{-3}$
50	1.232×10^{-2}	-1.154×10^{-2}	$+3.157 \times 10^{-3}$	-2.614×10^{-3}	$+1.330 \times 10^{-3}$

3. **The phase lags are smooth and never cross zero.** The unwrapped centreline lag between the antisymmetric velocity and thermal modes, $\Delta\varphi^{(a)}(0) = \arg\big(\widehat{V}^{(a)}/\Theta^{(a)}\big)$, increases monotonically from -2.154 to -1.532 rad over $Re \in [5, 50]$ and passes through no sign change; the symmetric-mode lag $\Delta\varphi^{(s)}(0)$ is non-monotonic. The local overlap factor of the sweep term stays close to unity ($0.993 \to 0.878$), i.e. that term remains nearly in phase, whereas the streamwise overlap factor oscillates about zero because of the nodes of the two modes. A narrative in which the reversal "coincides with a phase crossing" is not supported by these measurements; the correct statement is that the reversal coincides with the zero of the weighted overlap *sum* (83).

Physically, decreasing $Pr = Pe/Re$ at fixed Pe thins the thermal response relative to the velocity response, which weakens the transverse sweep production of the odd forcing faster than it weakens the streamwise-advection production. Once the two contributions become comparable, the small antisymmetric-velocity terms decide the sign, and the pumping direction reverses. This is the sense in which the dynamics, and not the symmetry, select the direction. Fig. 18 separates the two readings: the four additive contributions on top, and on the bottom the unwrapped phase lags, which cross zero nowhere and so rule out a phase mechanism.

9.2 Physical realization of the symmetry-selected pumping mechanism

The exact analysis above identifies the symmetry class of the nonlinear wall-flux redistribution, but the corresponding physical transport pathway is most clearly understood from the resolved two-dimensional reference solution. We therefore consider the representative parameter set

$$(k, Pe, Re, Pr, Gr) = (0.8, 7, 10, 0.7, 100), \tag{84}$$

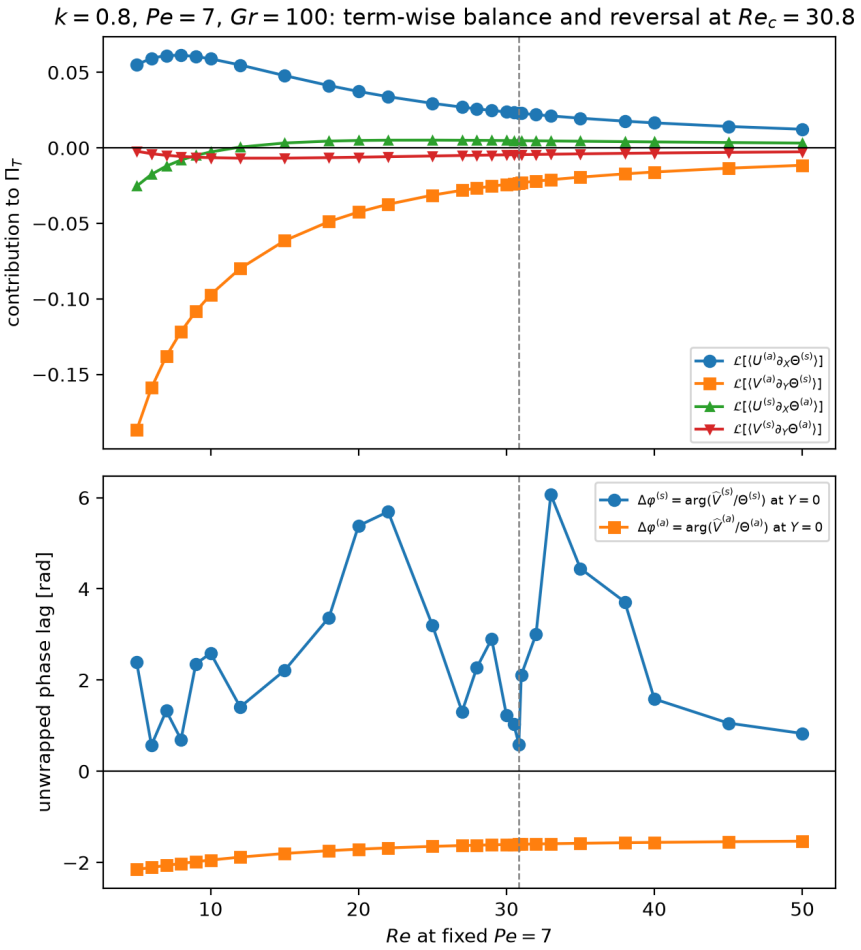


Figure 18: Upper panel: the four additive contributions (83) to Π_T versus Re at $k = 0.8$, $Pe = 7$; the dashed line marks $Re_c = 30.838$. Lower panel: unwrapped centreline phase lags, which vary smoothly and cross zero nowhere, showing that the reversal is a balance effect rather than a phase-sign effect.

which lies well inside the finite parameter domain investigated in the present study. The wavelength is neither in the asymptotic long-wave limit nor in the short-wave regime, while the buoyancy forcing is sufficiently strong for the secondary hydrodynamic response to be clearly resolved. This case is used throughout this section as a physical reference state.

A central point of the present interpretation is that the nonlinear pumping cannot be inferred from the total velocity field alone. The base Poiseuille flow remains dominant, whereas the buoyancy-induced perturbation is much smaller in amplitude but is responsible for the transverse transport that enters the quadratic thermal interaction. The reference solution therefore provides a direct physical realization of the analytical sequence

$$\theta_1 \longrightarrow \partial_Y\theta_1 \longrightarrow (U_1, V_1) \longrightarrow \mathcal{E}_{as} \longrightarrow \Pi_T. \tag{85}$$

9.2.1 Two-dimensional thermal structure and buoyancy-induced secondary motion

Figure 19 shows the first-order temperature response for the reference configuration together with its transverse gradient. The temperature field exhibits the imposed streamwise periodicity, while the transverse gradient is strongly structured across the channel. This distinction is important because the buoyancy term in the linearized momentum equation is proportional to $\Theta'(Y)$, rather than to the temperature amplitude itself. The resolved temperature field therefore provides the spatial structure of the thermal forcing that is subsequently converted into a hydrodynamic response.

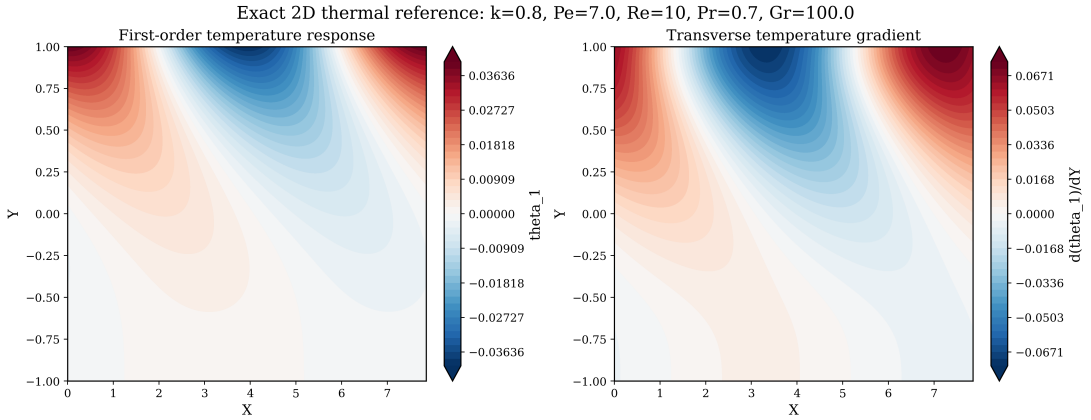


Figure 19: Two-dimensional thermal reference solution for $(k, Pe, Re, Pr, Gr) = (0.8, 7, 10, 0.7, 100)$. Left: first-order temperature response $\theta_1(X, Y)$. Right: transverse gradient $\partial_Y \theta_1$, which provides the leading buoyancy forcing of the hydrodynamic perturbation through Eq. (18).

The corresponding hydrodynamic response is shown in Fig. 20. The total streamwise velocity remains close to the Poiseuille reference profile,

$$U_0(Y) = \frac{3}{2}(1 - Y^2), \tag{86}$$

which confirms that the perturbation is a weak correction to the underlying forced flow. In the reference calculation,

$$\frac{\max |U_1|}{\max |U_0|} \simeq 6.35 \times 10^{-3}, \tag{87}$$

whereas the transverse velocity is of the same order as the perturbation streamwise velocity and is therefore dynamically relevant even though both remain small compared with the base flow.

For this reason, the streamlines shown in the right panel of Fig. 20 are constructed from the perturbation field (U_1, V_1) rather than from the total velocity $(U_0 + U_1, V_1)$. The latter would be overwhelmingly dominated by the nearly parallel Poiseuille component and

would obscure the secondary transport. The perturbation streamlines reveal a streamwise-periodic secondary circulation induced by buoyancy. These secondary motions provide the transverse advection required to convert the imposed thermal modulation into a nonzero quadratic mean response.

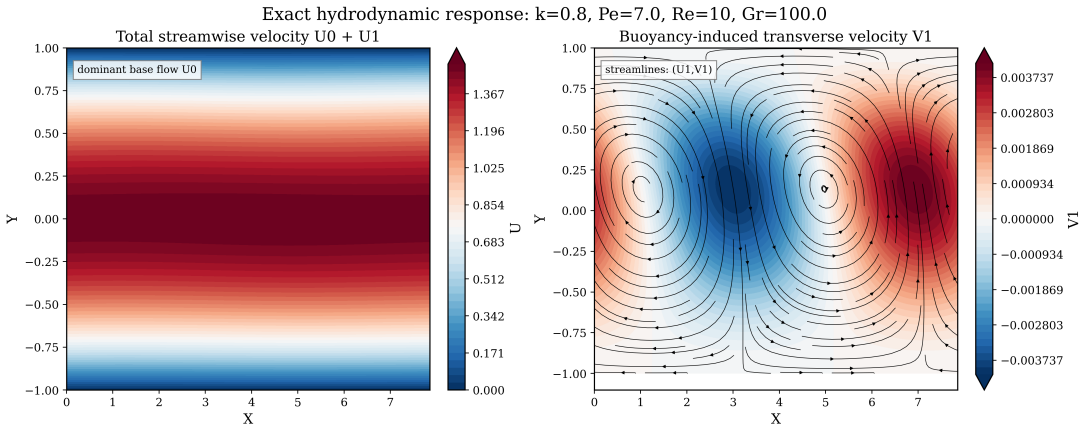


Figure 20: Hydrodynamic response for the reference case $(k, Pe, Re, Pr, Gr) = (0.8, 7, 10, 0.7, 100)$. Left: total streamwise velocity $U_0 + U_1$, showing the dominance of the Poiseuille base state. Right: buoyancy-induced transverse velocity V_1 with streamlines of the perturbation field (U_1, V_1). The secondary circulation, although small in amplitude, is the hydrodynamic carrier of the nonlinear thermal redistribution.

The reference solution therefore gives a useful physical interpretation of the analytical scaling

$$\widehat{V}^{(j)} \propto Gr. \tag{88}$$

Buoyancy does not alter the first-order thermal modes at fixed (k, Pe); instead, it provides the hydrodynamic mediator through which the thermal asymmetry is converted into a transverse transport process. This is also consistent with the exact switch-off result $\Pi_T = 0$ at $Gr = 0$. The profile-level signature of that odd response, for a strongly asymmetric wall forcing, is shown in Fig. 21.

9.2.2 Symmetry selection of the nonlinear interaction

The next step is to identify which part of the quadratic interaction is capable of producing a net wall-flux redistribution. Figure 22 shows the three quadratic forcing classes appearing in the streamwise-mean thermal equation. The self-interactions, $\mathcal{E}_{ss}$ and $\mathcal{E}_{aa}$, belong to the even symmetry class, whereas the mixed interaction $\mathcal{E}_{as}$ is odd under the reflection $Y \to -Y$.

The numerical structure shown in Fig. 22 provides a direct physical counterpart to the operator-based parity proof. The relevant observation is not simply that $\mathcal{E}_{as}$ changes sign across the channel, but that its positive and negative regions are organized in such a

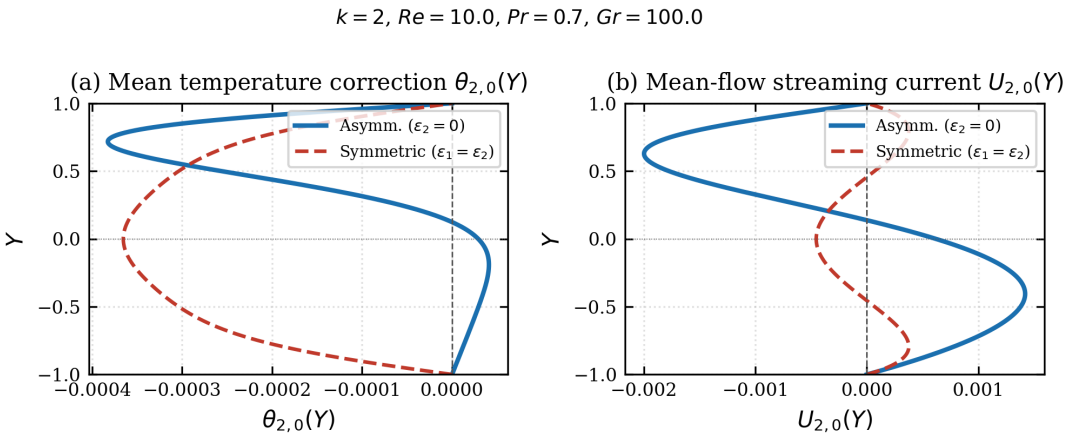


Figure 21: Cross-channel profiles of the mean temperature correction $\theta_{2,0}(Y)$ and the mean streaming velocity $U_{2,0}(Y)$ for a strongly asymmetric wall forcing. The odd thermal profile is the direct profile-level manifestation of the analytical symmetry theorem.

way that their unweighted integral cancels. Consequently, the mixed interaction does not act as a net volumetric heat source. Instead, it redistributes the mean thermal response transversely.

This distinction is essential. The nonlinear pumping observable is not controlled by the zeroth moment

$$\int_{-1}^{1} \mathcal{E}_{as}(Y)\, dY, \tag{89}$$

which vanishes, but by its first transverse moment. From Eq. (41),

$$\Pi_T = -\frac{Pe}{2} \int_{-1}^{1} Y\, \mathcal{E}_{as}(Y)\, dY. \tag{90}$$

The pumping mechanism can therefore be interpreted as a *transverse rectification* of a locally balanced nonlinear interaction: the mixed forcing produces no net mean heat source, but its spatial distribution with respect to the channel centerline generates a finite redistribution of the two wall heat fluxes.

This interpretation also explains why the distinction between symmetry and magnitude must be maintained. The odd parity of $\mathcal{E}_{as}$ is fixed by reflection symmetry, whereas the magnitude and sign of its first moment remain determined by the coupled thermal–hydrodynamic dynamics.

9.2.3 From local nonlinear interaction to wall-flux redistribution

The macroscopic consequence of the mixed interaction is the antisymmetric wall-flux correction established analytically above. For the fixed-Pe Reynolds-number comparison, Pr is adjusted according to $Pr = Pe/Re$; thus the cases $Re = 10$ and $Re = 35$ at $Pe = 7$

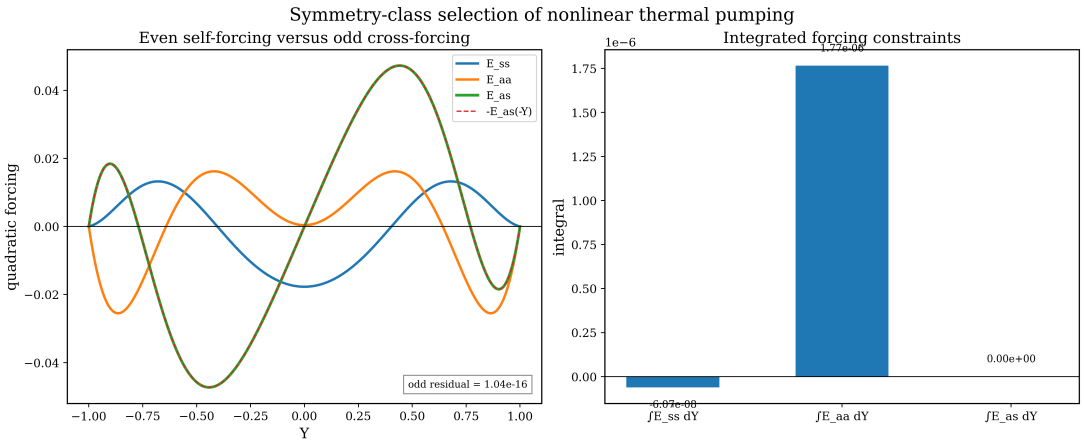


Figure 22: Symmetry classification of the quadratic thermal forcing for the reference configuration. Left: symmetric self-interactions $\mathcal{E}_{ss}$ and $\mathcal{E}_{aa}$ and mixed interaction $\mathcal{E}_{as}$. The latter satisfies $\mathcal{E}_{as}(-Y) = -\mathcal{E}_{as}(Y)$, with the displayed numerical parity residual of order 10^{-16}. Right: integrated forcing constraints. In particular, the mixed contribution has vanishing net integral, as required by its odd parity.

correspond to $Pr = 0.7$ and $Pr = 0.2$, respectively. For the present reference configuration ($Re = 10$, $Pr = 0.7$), the normalized pumping coefficient is

$$\Pi_T(Re = 10) = -4.7492 \times 10^{-2}, \tag{91}$$

while at the same k and Pe, but with $Re = 35$ ($Pr = 0.2$),

$$\Pi_T(Re = 35) = 6.7256 \times 10^{-4}. \tag{92}$$

Figure 23 displays the corresponding upper- and lower-wall corrections.

For $Re = 10$, the negative value of Π_T indicates that the upper-wall correction is directed opposite to the convention adopted for positive pumping, while the lower wall receives the compensating correction. The important point is that the total mean wall contribution remains balanced: the nonlinear mechanism redistributes the flux rather than creating an additional net heat input at second order.

The contrast between $Re = 10$ and $Re = 35$ is particularly informative. The first-order thermal problem depends on Re and Pr through the combination $Pe = RePr$, but the hydrodynamic response retains an explicit dependence on Re. Consequently, keeping Pe fixed does not preserve the nonlinear pumping response. The two cases therefore provide a direct physical demonstration of the parameter-dependence argument developed in Sec. 6: Reynolds number controls the hydrodynamic realization of the same thermal forcing.

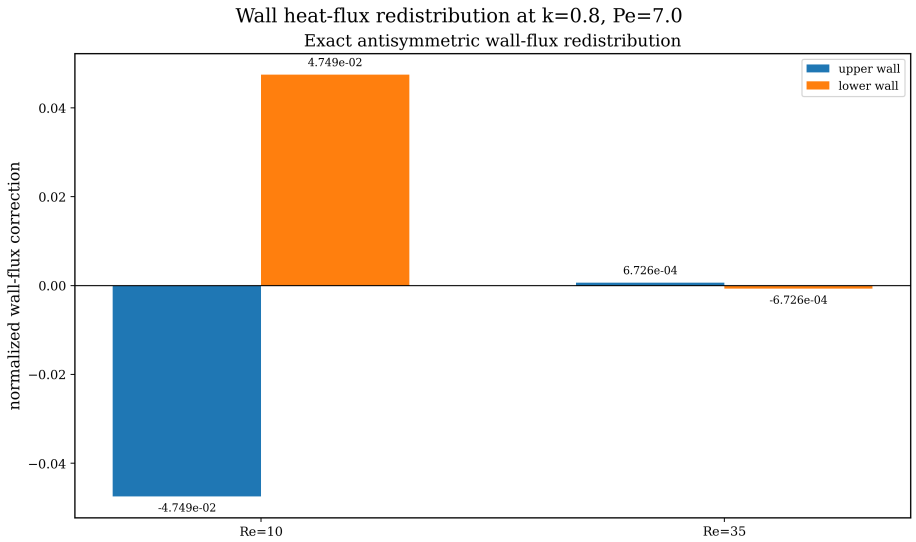


Figure 23: Mean nonlinear wall-flux redistribution for $k = 0.8$ and $Pe = 7$. The comparison is performed at fixed $Pe = 7$, with $Pr = Pe/Re$ adjusted accordingly. The upper and lower wall corrections are equal in magnitude and opposite in sign, in agreement with the exact relation $Nu_{2,0}(+1) = -Nu_{2,0}(-1)$. At $Re = 10$ the redistribution is comparatively strong and corresponds to $\Pi_T < 0$, whereas at $Re = 35$ the amplitude is much smaller and the sign has reversed.

9.2.4 Hydrodynamic control of pumping direction

The sign reversal observed in Fig. 23 is the most useful physical consequence of the coupled formulation because it goes beyond the exact symmetry statement. At fixed $k = 0.8$, $Pe = 7$ and $Gr = 100$, the normalized pumping response changes sign between $Re = 10$ and $Re = 35$, with a reversal near

$$Re_c \simeq 30.84. \tag{93}$$

This behavior should not be interpreted as a change in the symmetry class. The relation

$$Nu_{2,0}(+1) = -Nu_{2,0}(-1) \tag{94}$$

remains valid on both sides of the crossing. What changes is the sign of the first transverse moment in Eq. (90). Increasing Re modifies the phase and spatial distribution of the buoyancy-induced transverse velocity relative to the thermal field, and therefore changes the signed overlap that enters $\mathcal{E}_{as}$. The reversal is thus a *dynamical* effect superimposed on an unchanged symmetry constraint.

In physical terms, the imposed thermal modulation establishes the spatial temperature pattern, buoyancy converts the transverse temperature gradient into a secondary circulation, and the Reynolds number controls how that circulation is phase-shifted and spatially weighted relative to the thermal pattern. The resulting quadratic overlap can

consequently transfer the preferential wall-flux redistribution from one wall to the other without any change in the underlying reflection symmetry.

This provides a useful distinction between two classes of results obtained in the present study:

$$\boxed{\begin{array}{ll} \text{symmetry-controlled:} & Nu_{2,0}(+1) = -Nu_{2,0}(-1), \\ \text{dynamics-controlled:} & |\Pi_T|, \quad \mathrm{sign}(\Pi_T), \quad Re_c, \quad k^*(Pe). \end{array}} \tag{95}$$

The first relation is exact under the assumptions of the theory, whereas the second set of quantities must be determined from the coupled thermal and hydrodynamic response.

9.2.5 Physical picture of nonlinear thermal pumping

The resolved reference solution suggests a compact physical picture of the pumping mechanism. The periodic thermal boundary condition first creates a streamwise-organized temperature field. Its transverse gradient generates a buoyancy force, which produces a weak but spatially structured secondary velocity field. The interaction between the symmetric and antisymmetric components of this response generates the odd forcing $\mathcal{E}_{as}$. Although that forcing has zero net integral, its nonzero first transverse moment produces a finite wall-flux difference. The nonlinear pumping coefficient therefore measures a rectified transverse redistribution rather than a net creation of thermal energy.

The mechanism may be summarized schematically as

$$\begin{aligned} \text{periodic thermal pattern} &\xrightarrow{\partial_Y \theta_1} \text{buoyancy-induced secondary flow} \\ &\xrightarrow{\text{quadratic interaction}} \mathcal{E}_{as}(Y) \xrightarrow{Y\text{-moment}} \Pi_T. \end{aligned}$$

This interpretation also clarifies why the pumping disappears in the forced-convection limit $Gr = 0$. Without buoyancy, the secondary hydrodynamic mediator vanishes and the mixed thermal interaction cannot generate the wall-flux redistribution considered here. Conversely, when buoyancy is present, the symmetry-breaking boundary condition determines the admissible redistribution pattern, while the hydrodynamic parameters determine its magnitude and direction.

A useful physical interpretation follows from the geometry of the zero set. The disappearance of positive pumping for $Pe > Pe_c$ does not occur through a loss of the symmetry-breaking mechanism itself. The odd cross-interaction $\mathcal{E}_{as}$ remains structurally admissible, and the exact wall-flux antisymmetry remains unchanged. Instead, the region of parameter space in which the first transverse moment of $\mathcal{E}_{as}$ is positive contracts toward larger wavenumbers until the zero set reaches the upper boundary $k = 10$. The post-critical regime therefore corresponds to the expulsion of the sign-changing locus from the finite admissible domain, rather than to the disappearance of the underlying nonlinear interaction. This interpretation is consistent with the simultaneous preservation of the symmetry

class and the continuous local sign change observed in $\Pi_T(10, Pe)$ near Pe_c.

9.3 Buoyancy as the hydrodynamic mediator

The mechanism may be viewed as a two-stage process. The boundary asymmetry creates simultaneous symmetric and antisymmetric thermal modes. Buoyancy converts these temperature modes into transverse velocity modes, and the resulting velocity–temperature products generate the odd second-order forcing. The boundary symmetry therefore selects the admissible structure of the response, while buoyancy controls its magnitude through the linear factor Gr. Table 10 states the contrast with the symmetry-preserving forcing term by term.

Table 10: Symmetry-based comparison of the leading nonlinear transport response.

Property	Symmetry-preserving forcing	Asymmetric forcing
Wall parity	even or purely odd	mixed s–a content
Leading mixed forcing	absent	odd $\mathcal{E}_{as}$
Mean mixed temperature	symmetry-preserving	odd in Y
Wall-flux difference	zero	generally nonzero
Leading consequence	no antisymmetric redistribution	transverse thermal pumping

9.4 Spatial rectification and parameter-space structure

The reference solution supports an interpretation of nonlinear thermal pumping as a spatial rectification mechanism. In conventional thermal transport, a periodically forced field may redistribute heat locally while leaving the streamwise average unchanged. Here, however, the combination of boundary asymmetry, buoyancy-induced transverse transport, and quadratic mode coupling produces a directed *transverse* redistribution between the two walls.

The mathematical structure reveals why this effect cannot be reduced to a simple increase or decrease of the total heat transfer. The mixed forcing is odd and therefore has zero zeroth moment,

$$\int_{-1}^{1} \mathcal{E}_{as}(Y)\, dY = 0, \tag{96}$$

but its first moment is generally nonzero,

$$\int_{-1}^{1} Y\, \mathcal{E}_{as}(Y)\, dY \neq 0. \tag{97}$$

This distinction identifies the pumping mechanism as a redistribution of nonlinear transport rather than a generation of net mean thermal energy. The channel therefore acts as a transverse spatial rectifier: the zeroth moment cancels by symmetry, while the first

transverse moment survives and is converted directly into a measurable wall-flux difference.

The Reynolds-number reversal provides an additional indication that the effect is fundamentally a coupled transport phenomenon. At fixed Pe, the thermal forcing remains in the same thermal control class, whereas the hydrodynamic response changes with Re. The sign reversal near $Re_c \simeq 30.84$ consequently demonstrates that the direction of the rectified wall-flux redistribution is selected by the phase and spatial structure of the buoyancy-mediated transport, not by the thermal forcing alone.

This viewpoint is distinct from the exact symmetry theorem. The theorem determines *which wall-flux combinations are admissible*; the coupled transport problem determines *how strongly and in which direction* the admissible redistribution occurs.

The physical reference case also provides a useful interpretation of the broader (k, Pe, Re) parameter-space results. The long-wavelength result

$$A(k, Re, Pr) = A_2(Re, Pr)k^2 + o(k^2) \tag{98}$$

shows that an increasingly long wall pattern suppresses the transverse rectification. This is consistent with the physical picture above: as the streamwise modulation becomes slowly varying, the secondary thermal and hydrodynamic fields lose the spatial variation required to generate a finite rectified first moment.

At the opposite end of the parameter range, the zero-crossing branch $k^*(Pe)$ reflects a change in the signed first moment of the odd mixed interaction. Its finite-domain endpoint at

$$Pe_c = 10.871940554611, \qquad k = 10 \tag{99}$$

is therefore naturally interpreted as the point at which the zero set leaves the finite computational domain, rather than as a universal instability threshold. In both the Reynolds-number reversal and the finite-domain transition, the symmetry class remains unchanged while the coupled dynamics select the sign and magnitude of the first moment.

The two-dimensional reference solution thus provides a direct physical realization of the analytical mechanism: the imposed periodic temperature modulation generates a spatially structured thermal gradient, buoyancy converts it into a weak but structured secondary circulation, and the quadratic interaction of the symmetric and antisymmetric components produces the odd forcing $\mathcal{E}_{as}$. Because that forcing has zero net integral but nonzero first transverse moment, the result is a redistribution of wall heat flux rather than a net heat-source correction. The nonlinear thermal pumping is therefore not an isolated numerical feature of a particular parameter set, but a structurally constrained mode interaction whose amplitude and direction are selected by the coupled thermal and hydrodynamic transport operators. The separation between exact symmetry selection and dynamical response is the principal physical interpretation of the mechanism established

here.

9.5 What is exact and what is dynamical?

Three different statements must be distinguished. First, the relation $Nu_{2,0}(+1) = -Nu_{2,0}(-1)$ is an exact consequence of reflection symmetry, uniqueness, and the mixed symmetric–antisymmetric interaction. Second, the magnitude and sign of Π_T are not fixed by symmetry and depend on the coupled thermal–hydrodynamic dynamics. Third, the reported $Pe_c = 10.871940554611$ is not a universal critical Péclet number: it is the value at which the observed zero set reaches the imposed finite boundary $k = 10$ for the particular fixed-Reynolds-number problem considered here.

The dense CPU calculation strengthens this distinction. Below the endpoint, an interior zero exists; at the endpoint, the zero is located at the boundary; above the endpoint, no positive response remains in the investigated wavenumber interval. Thus, the transition concerns the geometry of the zero set in a finite parameter domain rather than the onset of a new symmetry class or a hydrodynamic instability.

9.6 Limitations

The analysis is restricted to the Boussinesq approximation, a two-dimensional laminar base state, a single spatial Fourier harmonic, and small modulation amplitudes $\varepsilon \ll 1$ for which the second-order expansion is asymptotically consistent. The finite-domain statement for Pe_c is also explicitly restricted to $0.02 \leq k \leq 10$ and $Re = 10$, $Gr = 100$. Direct numerical simulation of the fully nonlinear equations would be needed to quantify finite-amplitude corrections, higher harmonics, and possible secondary instabilities outside the present perturbative regime.

9.7 Generalization to phase-shifted wall forcing

The phase-shifted forcing family is no longer a speculative extension: it is treated in full in Section 5, where it is shown to introduce a second, independent pumping coefficient $\Pi_T^{\perp}$ and to reproduce a pure $\sin\phi$ dependence for equal wall amplitudes (Corollary 2). The qualitative argument given here in earlier drafts—that the mixed product merely "suggests" $\Pi_T \propto \sin\phi$—is superseded by Proposition 2.

9.8 Fully nonlinear DNS confirmation

The perturbative campaign is complete and every claim in Sections 5–8 is backed by the second-order solver. The thirteen DNS runs designed to test the same claims nonlinearly were executed with the mesh, time step and integration length already validated in Table 5 ($N_x = 64$, $N_y = 49$, $\Delta t = 1.5 \times 10^{-3}$, 3×10^4 steps): 90–144 s per case, 24.8 min in total,

with final residuals below 2.2×10^{-8}. Each entry of Table 11 is therefore a measured flux difference compared with the two-coefficient phase law evaluated at the stated wall amplitudes,

$$q_+ - q_- = 2Nu_{2,0}(+1) = 2\big(c\,\Pi_T + d\,\Pi_T^{\perp}\big), \qquad c = \Re\big(\widehat{\varepsilon}_a\widehat{\varepsilon}_s^{*}\big), \quad d = \Im\big(\widehat{\varepsilon}_a\widehat{\varepsilon}_s^{*}\big), \tag{100}$$

with $\widehat{\varepsilon}_s = \frac{1}{2}(\varepsilon_1 + \varepsilon_2 e^{-i\phi})$ and $\widehat{\varepsilon}_a = \frac{1}{2}(\varepsilon_1 - \varepsilon_2 e^{-i\phi})$, and with $\Pi_T(k, Re)$, $\Pi_T^{\perp}(k, Re)$ from the same authoritative solver.

Table 11: DNS measurements against the two-coefficient prediction ($Pe = 7$, $Gr = 100$, $Pr = Pe/Re$). The first two result columns are wall-flux differences, $q_+ - q_-$; the last column is the same comparison divided by the bilinear factor and expressed in coefficient space, $|\Pi_T^{\mathrm{DNS}} - \Pi_T^{\mathrm{pred}}| = |\Delta(q_+ - q_-)|/(2|\widehat{\varepsilon}_a\widehat{\varepsilon}_s^{*}|)$. The conversion factor is $2|\widehat{\varepsilon}_a\widehat{\varepsilon}_s^{*}| = 3.0 \times 10^{-4}$ for the D3, D3b and D4 families, 1.2×10^{-3} for the ladder run D3L and 9.0×10^{-4} for the D5 quadrature runs; the ladder flux gap of 3.25×10^{-7} is therefore the coefficient-space deviation 2.71×10^{-4} quoted for it. The parity run has $\widehat{\varepsilon}_a = 0$ and no coefficient-space statement is possible. The bound column is the deviation model $\delta_\Pi = \max(2.71 \times 10^{-4}, 1.52\%\,|\Pi_T|)$ pre-registered from the completed fixed-Pe sweep, and $\leq$ marks where it holds. The single measured sign that differs from the prediction, marked [a], is the ladder run discussed in item 3 of Section 9.8.

case	purpose	Re	Pr	ϕ	$q_+ - q_-$ predicted	measured	dev. in Π
D3	bracket at $k = 0.8$	20	0.3500	0	-1.928365×10^{-6}	-1.799329×10^{-6}	4.30×10
D3	bracket at $k = 0.8$	40	0.1750	0	$+3.242275 \times 10^{-7}$	$+3.809729 \times 10^{-7}$	1.89×10
D3	bracket at $k = 1.0$	15	0.4667	0	-1.828239×10^{-6}	-1.604171×10^{-6}	7.47×10
D3	bracket at $k = 1.0$	25	0.2800	0	$+4.671537 \times 10^{-7}$	$+5.910363 \times 10^{-7}$	4.13×10
D3	bracket at $k = 1.2$	10	0.7000	0	-3.702864×10^{-6}	-3.326895×10^{-6}	1.25×10
D3	bracket at $k = 1.2$	20	0.3500	0	$+7.719747 \times 10^{-7}$	$+9.454173 \times 10^{-7}$	5.78×10
D3	bracket at $k = 0.6$	40	0.1750	0	-3.886322×10^{-7}	-3.542744×10^{-7}	1.15×10
D3	bracket at $k = 0.6$	70	0.1000	0	$+1.740707 \times 10^{-7}$	$+1.910334 \times 10^{-7}$	5.66×10
D3L	crossing ladder	30	0.2333	0	-2.350614×10^{-7}	$+9.007819 \times 10^{-8}$ [a]	2.71×10
D4	control, far side	50	0.1400	0	$+3.989744 \times 10^{-7}$	$+4.416458 \times 10^{-7}$	1.42×10
D5	parity cancellation	10	0.7000	0	exactly 0	$+3.14 \times 10^{-18}$	—
D5	quadrature pumping	10	0.7000	$+\pi/2$	$+8.005465 \times 10^{-5}$	$+8.115924 \times 10^{-5}$	1.23×10
D5	reversed quadrature	10	0.7000	$-\pi/2$	-8.005465×10^{-5}	-8.115932×10^{-5}	1.23×10

The forcing amplitudes are $\varepsilon_1 = \varepsilon_s + \varepsilon_a$ and $\varepsilon_2 = \varepsilon_s - \varepsilon_a$ with $\varepsilon_s = 0.03$ and $\varepsilon_a = 0.005$ throughout, except for the ladder run D3L, which raises ε_a to 0.02 at fixed ε_s, and for the D5 family, which uses equal amplitudes $\varepsilon_1 = \varepsilon_2 = 0.03$ with a phase lag ϕ.

1. *Reversal confirmed at four wavenumbers.* The locus $Re_c(k)$ of Table 6 is perturbative, so each wavenumber is tested on both sides of its own crossing rather than on a common Reynolds interval: at $k = 0.6$ the reversal sits at $Re = 51.25$, and a grid confined to $Re \leq 40$ would sample a single sign at that wavenumber and measure nothing about a reversal. All eight bracket runs return the predicted sign, and the two members of every pair return opposite signs. The reversal of the pumping

direction with Re at fixed Pe is therefore a nonlinear result, not an artefact of the second-order truncation.

2. *Control point away from the crossing.* The relative deviation of the earlier DNS comparison was largest near $Re = 35$, where Π_T approaches zero and a fixed absolute error maps onto a large relative one. At $Re = 50$ the measured coefficient is $+1.472\times 10^{-3}$ against $+1.330 \times 10^{-3}$ predicted: the sign is unambiguous and the absolute deviation, 1.42×10^{-4}, is half the pre-registered floor. The large relative error near the crossing is thus a crossing artefact.

3. *Error-model discriminator.* The ladder run D3L was designed to separate the two readings of δ_Π. Raising ε_a fourfold multiplies the flux gap, 3.25×10^{-7} against the 1.29×10^{-7} measured at the reference amplitude and the nearest bracket point of the same wavenumber ($k = 0.8$, $Re = 20$), but leaves the coefficient-space deviation at 2.709×10^{-4}—that is 1.00 times the absolute floor—and the measured sign is positive where the prediction is negative. The deviation is therefore *not* absolute in $q_+ - q_-$ and does not improve with the forcing amplitude: the coefficient-space reading is the correct one, and a single-amplitude DNS test of this family cannot resolve the reversal once $|\Pi_T| \lesssim \delta_\Pi$.

4. *Quadrature coefficient.* The equal-amplitude, phase-shifted runs measure $Nu_{2,0}(+1) = +4.0580 \times 10^{-5}$ at $\phi = \pi/2$ and -4.0580×10^{-5} at $\phi = -\pi/2$, against $+4.003 \times 10^{-5}$ and -4.003×10^{-5} predicted: a relative deviation of 1.38% and an antisymmetry residual of 8.0×10^{-11}, i.e. 9.9×10^{-7} of the signal. The existence and the sign reversal of $\Pi_T^{\perp}$ are thus confirmed nonlinearly, and with them Corollary 2. The $\phi = 0$ member of the same family is the exact cancellation limit: the two wall fluxes are measured equal to 3.1×10^{-18}, against individual values of $-1.145796985 \times 10^{-7}$, so the parity theorem is reproduced in DNS at a relative precision of 2.7×10^{-11}—eleven digits beyond the flux itself. This is the sharpest single confirmation in the campaign, and it is independent of the symmetric-forcing case of Table 5 at 8×10^{-18}.

Fig. 24 collects the two readings of the same campaign: the reversal direction, panel (a), and the magnitude against the pre-registered deviation model, panel (b), which is the quantity the analysis of Sec. 9.9 judges.

9.9 What the campaign changes about the error model

The campaign falsifies part of its own pre-registered error model. Of the twelve non-degenerate cases, seven satisfy $\delta_\Pi = \max(2.71 \times 10^{-4}, 1.52\%\,|\Pi_T|)$ and five exceed it, by up to 4.6 times. The pattern is systematic rather than random: the coefficient-space deviation is independent of the forcing amplitude (item 3 above) and grows monotonically

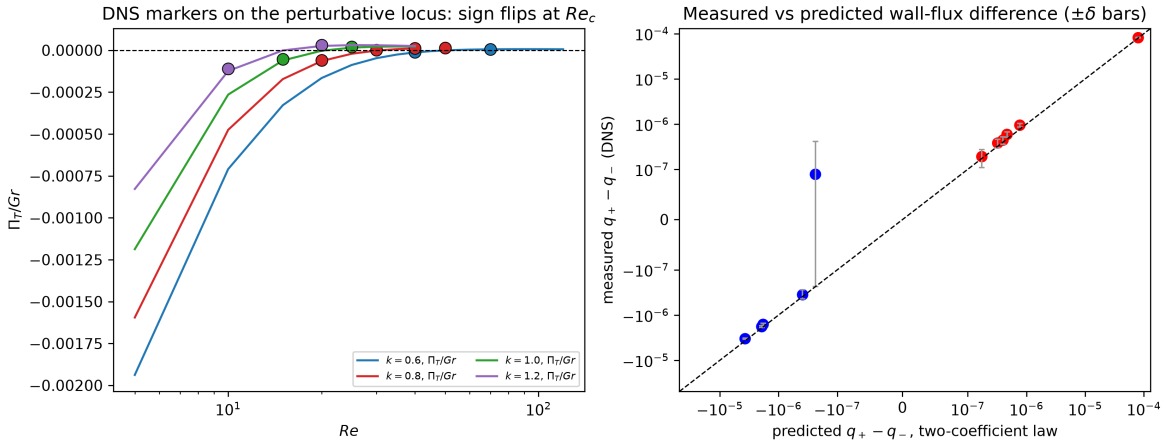


Figure 24: (a) Measured pumping coefficient divided by Gr (markers) against the perturbative locus (lines) at the four wavenumbers bracketed by DNS; every marker pair straddles its own Re_c with opposite sign. This panel tests the direction of the reversal. (b) Measured versus predicted wall-flux difference for the twelve non-degenerate runs, with error bars $2|\widehat{\varepsilon}_a\widehat{\varepsilon}_s^*|\delta_\Pi$. This panel tests the magnitude, and is the one the error model of Sec. 9.9 is judged against. Eleven of the twelve lie on the predicted side of zero; the outlier is the ladder run, the single case whose margin was below unity before the campaign was launched.

with the wavenumber, its maximum over the bracket pairs rising from 1.15×10^{-4} at $k = 0.6$ through 4.30×10^{-4} at $k = 0.8$ and 7.47×10^{-4} at $k = 1.0$ to 1.25×10^{-3} at $k = 1.2$. Two consequences follow. First, the 1.52% figure quoted for the reference case is specific to $(k, Re) = (0.8, 10)$, where $|\Pi_T|$ is large; near a crossing the relevant quantity is the absolute offset, not a relative one. Second, a DNS test of the reversal at wavenumber k is meaningful only for $|\Pi_T(k, Re)| \gtrsim \delta_\Pi(k)$, which the bracket design of item 1 satisfies in every pair except the ladder run.

9.10 What drives the growth of the remainder with k

The monotone growth is the interesting part of the table, and it has two candidate causes that this campaign does not separate. The first is physical: the second-order truncation neglects harmonics $2k$ and $3k$ and the wall-normal gradients grow with k, so a higher-order remainder is expected to increase with the wavenumber. The deviation grows monotonically with k over the four resolved wavenumbers, and a log–log fit of the eight bracket runs gives $|\Delta\Pi_T| \propto k^{3.4\pm1.7}$ at $R^2 = 0.84$ over the narrow span $k = 0.6$–1.2. That exponent is not statistically sharp enough to discriminate between candidate higher-order mechanisms, so the growth is reported as an empirical observation and not as a scaling law: no power of k enters any subsequent argument of this paper, and the separation of the growth from the mesh-resolution floor is identified as future work below. The second candidate is numerical, and it cannot be dismissed by the residuals alone. The time-stepping

and solver errors are genuinely negligible—the final residuals are below 2.2×10^{-8}, four orders under the deviations they accompany—but the spatial discretisation is not: on the published convergence ladder at $k = 0.8$, Π_T moves by 3.1×10^{-4} between the production mesh of 3136 degrees of freedom and the finest tested, 7008, which is the same order as the deviations measured here at $k = 0.8$ and $k = 1.0$. Because the mesh is held at $N_x = 64$ while the wavelength shrinks, a truncation error of this kind predicts precisely the observed growth with k.

Two observations narrow the field. The deviation is amplitude-independent at fixed k (item 3), whereas a genuine higher power of the forcing would scale with ε^2; the remainder is therefore either a mesh effect or a k-dependent contribution that does not multiply $\widehat{\varepsilon}_a \widehat{\varepsilon}_s^*$, such as the feedback of the second-order mean field on the base temperature gradient. And the growth is confined to the comparison with the perturbative coefficient: the *sign* structure, which is what the paper claims, is recovered in all eight bracket pairs. A decisive separation requires one replica of the $k = 1.2$ pair at refined N_x: if the deviation collapses toward the floor, the growth is discretisation; if it persists, it is the remainder of the truncation. That test is identified as future work rather than asserted here.

Grashof-independence of the reversal point (Corollary 1) is verified perturbatively to better than 10^{-9} in Re_c and now nonlinearly as well: DNS at the Grashof pair $(Gr, Re) = (50, 20)$, $(200, 20)$, $(50, 50)$, $(200, 50)$ returns Grashof ratios 4.0003 and 4.0015 with the predicted sign preserved in both pairs (Sec. 8.1).

10 Conclusions

This study separates what reflection symmetry forces from what the coupled thermal–hydrodynamic dynamics choose. The organizing result is that the *structure* of the nonlinear wall-flux redistribution is fixed exactly by symmetry, whereas its *amplitude and sign* are dynamical, and that the Reynolds number—even at fixed Péclet number—controls the direction of the resulting thermal pumping.

What symmetry selects. The first-order thermal fields decompose uniquely into symmetric and antisymmetric parts; their hydrodynamic responses inherit opposite reflection parities, so the mixed quadratic forcing is necessarily odd across the channel. The second-order mean temperature correction is therefore odd, which imposes

$$\boxed{Nu_{2,0}(+1) = -Nu_{2,0}(-1)} \tag{101}$$

This is a structural consequence of the governing equations: it holds whether or not the pumping amplitude is large, small, or zero.

What the dynamics select. The redistribution is measured by the thermal-pumping

coefficient, for which the Green identity $\Pi_T = -(Pe/2)\int_{-1}^{1} Y\,\mathcal{E}_{as}\,dY$ yields

$$\Pi_T = Gr\,A(k, Re, Pr), \qquad A(k, Re, Pr) = O(k^2) \;\text{ as }\; k \to 0. \tag{102}$$

Linearity in Gr separates the buoyancy amplitude from the response, and the long-wavelength limit shows that a slowly varying wall pattern cannot drive the effect. Because Π_T is proportional to Gr, its zero is independent of Gr (Corollary 1): the reversal point is $Re_c = 30.8384965$ for $Gr = 25, 50, 100$ and 200, with an absolute spread of 1.2×10^{-10} and A invariant to better than 10^{-10} relative. DNS reproduces the proportionality itself: at $Re = 20$ and $Re = 50$ the measured Grashof ratio of $q_+ - q_-$ is 4.0003 and 4.0015, and the sign predicted on either side of the crossing survives.

Phase selects a second, independent coefficient. Allowing a phase lag ϕ between the two walls makes the mixed product complex, and streamwise averaging leaves two independent invariants rather than one. The wall flux becomes

$$Nu_{2,0}(+1) = \Re(\varepsilon_a \varepsilon_s^*)\,\Pi_T + \Im(\varepsilon_a \varepsilon_s^*)\,\Pi_T^{\perp}, \tag{103}$$

with $\Pi_T = -4.74850846 \times 10^{-2}$ and $\Pi_T^{\perp} = +8.89496095 \times 10^{-2}$ at the reference state: the quadrature coefficient is of opposite sign and nearly twice as large. For equal wall amplitudes the in-phase invariant vanishes identically and the response obeys $Nu_{2,0}(+1) \propto \sin\phi$ exactly (Corollary 2), verified phase by phase to 2.4×10^{-13}. This is a selection rule, not a suggestion: it fixes which combinations of wall forcing can pump and in which quadrature.

Reversal is systematic, and it is a magnitude race, not a phase flip. The reversal point has been mapped over seven wavenumbers, giving Re_c from 79.08 at $k = 0.4$ to 8.54 at $k = 2.0$; the reversal is resolved at every wavenumber sampled, and its locus follows $Re_c \approx 21.50\,k^{-1.47}$ to within 11% over that range. The DNS confirms the reversal at the reference wavenumber: $\Pi_T^{\mathrm{DNS}} = -4.6763 \times 10^{-2}$, 1.52% from the perturbative -4.7485×10^{-2}, with the sign change between $Re = 20$ and $Re = 30$ at fixed $Pe = 7$ reproduced nonlinearly, and at three further wavenumbers: all eight runs that bracket Re_c at $k = 0.6$, 0.8, 1.0 and 1.2 return the predicted sign, and the quadrature coefficient is measured to 1.38% (Sec. 9.8). The term-wise decomposition of Π_T shows what does *not* happen: the antisymmetric wall-normal transport term never crosses zero and the unwrapped phase lag $\Delta\varphi^{(a)}$ increases monotonically from -2.154 to -1.532 rad across the reversal. The sign change arises because the magnitudes of the advective and wall-normal contributions become comparable ($|T_{Va}|/|T_{Ua}|$ falls from 3.40 to 0.936), with the subdominant symmetric terms deciding the crossing to within 6%.

Independent nonlinear evidence. DNS confirms the Grashof scaling $\Pi_T \propto Gr$ with a coefficient of determination effectively equal to unity and a vanishing response at $Gr = 0$, recovers the reflection identity $\Pi_T(\varepsilon_1, \varepsilon_2) = -\Pi_T(\varepsilon_2, \varepsilon_1)$, and reproduces the exact

cancellation of the parity limit: at equal wall amplitudes and $\phi = 0$ the two measured wall fluxes agree to 3.1×10^{-18}, against individual fluxes of 1.15×10^{-7}, a relative precision of 2.7×10^{-11}. On the Grashof pair of Sec. 8.1 the nonlinear correction is itself proportional to Gr, so it rescales the coefficient without shifting its zero: Corollary 1 survives beyond the perturbative regime. Spatial and temporal refinement demonstrate convergence of the nonlinear observable. The campaign also falsifies part of its own pre-registered error model: five of the twelve non-degenerate cases exceed δ_Π, with an excess that is independent of the forcing amplitude and grows with the wavenumber (Sec. 9.10); separating the truncation remainder from the residual spatial discretisation requires a refined-N_x replica of the $k = 1.2$ bracket pair, which is the immediate next test. The remaining finite-domain statement $Pe_c = 10.871940554611$ is the value at which the zero set of $\Pi_T(k, Pe)$ leaves the imposed interval $0.02 \le k \le 10$ at $Re = 10$; it is a property of the window, not an instability threshold.

Future work should extend the present approach beyond the two-dimensional laminar setting: larger modulation amplitudes, three-dimensional configurations and turbulent mixed convection. The most immediate testable consequence of the two-coefficient law is the quadrature prediction of Section 9.8—equal-amplitude forcing at $\phi = \pi/2$ gives $Nu_{2,0}(+1) = +4.0580\times10^{-5}$ in DNS against $+4.003\times10^{-5}$ predicted, whereas the in-phase-dominated forcing used in the earlier validation gives -7.123×10^{-6}; the two cases differ only in $\Im(\varepsilon_a\varepsilon_s^*)$. Wall-temperature phase and amplitude asymmetry therefore constitute a control knob for the direction of thermal pumping, with the Reynolds number setting where the knob changes sign.

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Data availability

All numerical data supporting this study are assembled in a supplementary package of 64 files, documented item by item and inventoried with the size, line count and SHA-256 digest of every entry. Section S1 holds the dense Pe survey behind Fig. 6; S2 the perturbative reversal and phase campaigns of Secs. 8 and 5; S3 the nonlinear DNS results of Sec. 7, including the pre-run predictions, the raw per-case measurements, the derived deviation table and the analysis summary of the confirmation campaign of Sec. 9.8, the four-run Grashof-pair matrix of Sec. 8.1 and the reduced data behind Fig. 17; and S4 the boundary-value, shooting and analysis scripts that regenerate every table and figure of

this paper from the raw records. The DNS matrix runs from a single driver that performs a dry run unless an explicit go-ahead flag is supplied and resumes from its own checkpoint. The package inventory maps each figure and table of the paper to the files that produce it.

A Derivation of the first-order hydrodynamic BVP

For the e^{-ikX} convention, incompressibility gives $-ik\widehat{U}+\widehat{V}'=0$, hence $\widehat{U}=-i\widehat{V}'/k$. The complex perturbation vorticity is

$$\widehat{\omega}=\partial_X\widehat{V}-\partial_Y\widehat{U}=\frac{i}{k}(D^2-k^2)\widehat{V}. \tag{104}$$

Taking the streamwise derivative of the transverse momentum equation minus the transverse derivative of the streamwise momentum equation yields

$$U_0\partial_X\omega_1-V_1U_0''=\frac{1}{Re}\nabla^2\omega_1-\frac{Gr}{Re^2}\partial_Y\theta_1. \tag{105}$$

Substitution of the modal representation gives Eq. (18). The no-slip conditions imply $\widehat{V}(\pm1)=\widehat{V}'(\pm1)=0$.

B Reflection-operator proof

Define $\mathcal{R}[f](Y)=f(-Y)$. Since $\mathcal{R}D=-D\mathcal{R}$, $\mathcal{R}D^2=D^2\mathcal{R}$, and U_0, U_0'' are even,

$$\mathcal{R}\mathcal{L}_T=\mathcal{L}_T\mathcal{R}, \qquad \mathcal{R}\mathcal{L}_H=\mathcal{L}_H\mathcal{R}, \tag{106}$$

where $\mathcal{L}_T$ and $\mathcal{L}_H$ denote the thermal and hydrodynamic operators. Uniqueness of the corresponding BVPs therefore fixes the parity classes of all first-order modal fields.

C Proof of the odd cross-interaction

Using Eq. (28), the four terms have parity

$$\widehat{U}^{(a)}\overline{-ik\Theta^{(s)}}:\text{odd}, \tag{107}$$
$$\widehat{U}^{(s)}\overline{-ik\Theta^{(a)}}:\text{odd}, \tag{108}$$
$$\widehat{V}^{(a)}\overline{(\Theta^{(s)})'}:\text{odd}, \tag{109}$$
$$\widehat{V}^{(s)}\overline{(\Theta^{(a)})'}:\text{odd}. \tag{110}$$

Therefore $\mathcal{E}_{as}$ is odd and its channel integral vanishes.

D Mean momentum balance and fixed-bulk-flow constraint

The mean second-order momentum equation is

$$\frac{1}{Re}U_{2,0}'' + G_2 = \langle U_1 \partial_X U_1 + V_1 \partial_Y U_1 \rangle_X - \frac{Gr}{Re^2}\theta_{2,0}. \tag{111}$$

The conditions $U_{2,0}(\pm 1) = 0$ and $\int_{-1}^{1} U_{2,0}\, dY = 0$ determine G_2 under the fixed-bulk-flow formulation. This streaming problem is not required for the proof of wall-flux antisymmetry, but it is needed for the reported mean-flow validation.

E Proof of the nonlinear thermal-pumping theorem

From

$$\theta_{as,2}'' = Pe\mathcal{E}_{as}, \qquad \theta_{as,2}(\pm 1) = 0, \tag{112}$$

and oddness of $\mathcal{E}_{as}$, the function $-\theta_{as,2}(-Y)$ satisfies the same BVP as $\theta_{as,2}$. Uniqueness gives $\theta_{as,2}(-Y) = -\theta_{as,2}(Y)$ and thus an even derivative. The inward-normal definition of the two wall fluxes then immediately yields $Nu_{2,0}(+1) = -Nu_{2,0}(-1)$.

F Proof of linearity in Grashof number

The thermal modes are independent of Gr, while the hydrodynamic BVP is linear with forcing proportional to $Gr\Theta'$. Consequently $\widehat{V}, \widehat{U}, \mathcal{E}_{as}$ and Π_T are each proportional to Gr at fixed (k, Re, Pr).

G Integral representation of the pumping coefficient

For $\theta'' = f(Y)$ with $\theta(\pm 1) = 0$,

$$\theta'(1) = \frac{1}{2}\int_{-1}^{1}(1+Y)f(Y)\, dY, \qquad \theta'(-1) = -\frac{1}{2}\int_{-1}^{1}(1-Y)f(Y)\, dY. \tag{113}$$

For odd $f = Pe\mathcal{E}_{as}$,

$$\theta'(1) = \theta'(-1) = \frac{Pe}{2}\int_{-1}^{1} Y\mathcal{E}_{as}(Y)\, dY, \tag{114}$$

and the inward-normal definition gives Eq. (41).

H Independent validation by initial-value shooting

For the thermal BVP, the unknown complex derivative at $Y = -1$ is adjusted to satisfy the prescribed value at $Y = +1$. For the fourth-order hydrodynamic BVP, the two complex initial derivatives $\widehat{V}''(-1)$ and $\widehat{V}'''(-1)$ are adjusted until $\widehat{V}(+1) = \widehat{V}'(+1) = 0$. The shooting solutions are compared against the adaptive collocation solutions, and Π_T is independently reconstructed from the first-order fields.

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